

ESSENTIAL MATHEMATICS GRADE 10: QUADRATIC EQUATIONS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	Essential Mathematics
Strand	Strand 1.0: Numbers and Algebra
Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Formation of algebraic expressions from real-life situations (using letters to represent unknown quantities) – Formation of quadratic expressions from real-life contexts (e.g. areas of rectangles with unknown sides, chairs arranged in rows and columns) – Formation of quadratic equations from real-life situations – Identifying and writing quadratic expressions in standard form ($ax^2 + bx + c$) – Factorisation of quadratic expressions (product-and-sum method) – Solving quadratic equations by the factor method (factorisation) – Verifying solutions of quadratic equations by substitution – Applications of quadratic expressions and equations in real-life situations (area of land, seating arrangements, farming layouts)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: - form algebraic expressions from different situations, - form quadratic expressions and equations from real life situations, - factorise quadratic expressions in different situations, - solve quadratic equations by factorisation, - recognise use of quadratic expressions in real life situations.
Core Competencies	– Communication and Collaboration: The learner collaborates with peers as they discuss and work out factors of quadratic expressions, share reasoning about real-life situations modelled by quadratic equations, and present findings to the class. – Critical Thinking and Problem Solving: The learner develops open-mindedness by interpreting real-life scenarios (e.g. farm plots, seating arrangements in a school hall in Nairobi), translating them into quadratic expressions, and systematically working out factors and solutions. – Digital Literacy: The learner uses digital resources or print media to search for applications of quadratic equations in real life, practising online safety measures while doing so.
Core Values	– Respect: The learner displays humility by listening to others' opinions while discussing areas where quadratic equations are used in real life situations. – Integrity: The learner is honest when verifying solutions to quadratic equations by substitution, and does not copy peers' workings without understanding.

	– Responsibility: The learner takes ownership of group tasks, contributes fairly to collaborative activities, and observes online safety procedures when searching for applications of quadratic equations using digital resources.
Science & Engineering Practices	– Asking Questions and Defining Problems: Learners identify real-life situations (e.g. a farmer in the Rift Valley fencing a rectangular shamba of known area) that can be modelled as quadratic equations, and frame these as mathematical questions to be solved. – Using Mathematics and Computational Thinking: Learners translate word problems into algebraic expressions and quadratic equations, apply the factorisation algorithm systematically, and verify answers through substitution — demonstrating mathematical reasoning as a core scientific practice. – Constructing Explanations and Designing Solutions: Learners explain each step of the factorisation process, justify why both solutions of a quadratic equation may or may not be valid in a real-life context, and communicate solutions clearly to peers. – Obtaining, Evaluating, and Communicating Information: Learners use digital resources and print media to research applications of quadratic equations, evaluate the reliability of sources, observe online safety, and share findings with the class.
Pertinent & Contemporary Issues (PCIs)	– Online Safety: The learner observes online safety procedures and guidelines while searching for applications of quadratic equations using digital resources (e.g. Kenya Education Cloud, educational websites). – Life Skills — Critical Thinking and Problem Solving: The learner applies logical reasoning to translate real-world Kenyan contexts (market stalls, farming plots, school seating layouts) into quadratic models and solve them systematically. – Financial Literacy: The learner recognises how quadratic relationships appear in commercial contexts such as profit-and-loss maximisation for small businesses and market traders, linking algebra to economic decision-making.
Career Connections	– Architecture and Construction: Quadratic equations are used to calculate areas and dimensions of rooms, buildings, and land plots — directly relevant to careers in construction and quantity surveying in Kenya. – Agriculture and Farm Management: Farmers in the Rift Valley and Central Kenya use quadratic relationships when planning rectangular plots to maximise yield for a given perimeter or area. – Business and Entrepreneurship: Quadratic models underpin profit maximisation and cost optimisation in sole proprietorships, market trading, and small business management. – Journalism and Media: Data journalists use algebraic modelling including quadratic relationships to analyse and present trends in population growth, sports statistics, and economic indicators. – Sports Management and Coaching: Quadratic equations describe projectile motion in athletics (e.g. a javelin thrown by a Kenyan athlete), relevant to sports science and coaching careers. – Engineering and Technology: Foundational algebraic skills including quadratic equations support further study in civil, electrical, and mechanical engineering at technical and tertiary institutions.
Focus for Lessons	This unit develops learners' ability to move fluently between real-life Kenyan contexts and algebraic representations, beginning with forming algebraic and quadratic expressions, progressing to factorisation, and culminating in solving quadratic equations and interpreting solutions in context. The pedagogical approach is collaborative and problem-based: learners encounter authentic scenarios — such as a farmer in Kericho planning a rectangular tea plot, or a school in Kisumu arranging chairs in rows and columns — and use these to build meaning for quadratic structures before applying formal procedures. Throughout the unit, digital tools and print resources are used to deepen understanding and explore real-world applications, with online safety observed at all times.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya? KICD KEY INQUIRY QUESTION:

	1. How are quadratic equations applied in real life?
Anchoring Phenomenon	A school in Kisumu is preparing for its annual prize-giving ceremony. The administration needs to arrange a rectangular block of chairs for 120 guests so that the number of columns exceeds the number of rows by 7. A student volunteer, Achieng, sketches the layout on paper and realises she cannot simply divide 120 by a number — she needs to find two related numbers whose product is 120. When she tries to "guess and check," she keeps getting decimals or the wrong totals. Her teacher, Mr Otieno, suggests that the relationship between the rows and columns can be written as an equation involving a squared unknown. Students find this puzzling: why does a simple seating problem produce an equation with a squared term? How can you solve an equation where the unknown appears multiplied by itself? And once you find two answers, which one actually works for the chairs — or do both?
Storyline Thread	Lesson 1 — Anchoring Phenomenon & Forming Algebraic Expressions: Learners are introduced to Achieng's chair-arrangement problem at the Kisumu prize-giving ceremony. They brainstorm unknown quantities in everyday Kenyan contexts (chairs, farm plots, market stalls), practise using letters to represent unknowns, and form simple algebraic expressions from word statements. The Driving Question Board (DQB) is introduced: learners post initial questions and wonderings about how algebra can help solve Achieng's problem. Lesson 2 — From Expressions to Quadratic Expressions: Learners revisit the chair problem and realise the product of two linear expressions produces a squared term. Guided by the teacher, they form quadratic expressions from contexts such as a rectangular maize plot in Eldoret (rows $\times$ columns = area) and the mandazi dough scenario. They identify the standard form $ax^2 + bx + c$ and distinguish quadratic expressions from linear ones. DQB is updated with new observations. Lesson 3 — Forming Quadratic Equations from Real-Life Situations: Learners read and interpret case scenarios (farm fencing in the Rift Valley, seating arrangements, seedling grids) and practise translating each into a quadratic equation. They discuss with peers, compare equations formed, and identify that a quadratic equation has exactly one squared term and equals a constant. Misconceptions are surfaced and addressed. DQB questions about "how to solve" are flagged for upcoming lessons. Lesson 4 — Factorisation of Quadratic Expressions (Product-and-Sum Method): Learners are explicitly taught the product-and-sum strategy for factorising quadratic expressions. They work through expressions derived from their real-life contexts (e.g. the number of seedlings: $n^2 + 3n - 108$). Peer discussion and collaborative practice reinforce the method. Learners verify factorisations by expanding. DQB is updated: "Can factorisation always work?" Lesson 5 — Solving Quadratic Equations by Factorisation: Learners apply factorisation to solve quadratic equations, setting each factor equal to zero and finding both roots. They return to Achieng's chair problem and the farm-fencing scenario, compute both solutions, and interpret which solution(s) make sense in context (rejecting negative values where appropriate). The teacher facilitates a whole-class sensemaking discussion connecting solutions back to the phenomenon. Lesson 6 — Verification and Deeper Factorisation Practice: Learners practise verifying solutions by substitution into the original equation. They tackle a range of quadratic equations of increasing complexity drawn from Kenyan contexts (matatu profit, market trading, athletics projectile scenarios). Peer checking and error analysis are used. Learners update the DQB, answering earlier questions about when both, one, or neither solution is valid in context. Lesson 7 — Applications and Digital Research: Learners use digital devices (or print resources) to research real-world applications of quadratic equations — in architecture, agriculture, sports, and business — observing online safety measures. Small groups prepare short presentations linking their findings to the driving question. The teacher facilitates a gallery walk and class discussion. Learners add new connections to the DQB and revise their models of "what a quadratic equation can tell us."

	Lesson 8 — Model Building, Consolidation & Assessment: Learners construct a cumulative visual model (poster or digital slide) that maps the full journey: real-life context → algebraic expression → quadratic expression → quadratic equation → factorisation → solution → interpretation in context. They present their models, respond to driving question prompts, and complete a class activity / written assessment task. The DQB is reviewed as a class — all initial questions are answered or noted for further inquiry. Lesson closes with learners articulating how quadratic equations are used across careers in Kenya.
Supporting Phenomena	– A farmer in the Rift Valley has 56 metres of fencing and wants to enclose a rectangular shamba whose length is 2 metres more than twice its width — learners discover that finding the dimensions requires solving a quadratic equation, and that one solution may be negative (physically meaningless) while the other gives the actual farm dimensions. – A mandazi vendor in Mombasa's Old Town cuts a square piece of dough and removes a smaller square from one corner to create an L-shaped piece; knowing the total area of dough remaining, learners write and solve a quadratic equation to find the side length of the original square. – A matatu operator in Nairobi observes that his daily profit (in Kenya Shillings) depends on the number of trips made, modelled by an expression with a squared term; learners form and factorise the quadratic expression to find the number of trips that gives zero profit (break-even), connecting algebra to commercial arithmetic. – A teacher at a school in Eldoret plants maize seedlings in a rectangular grid where the total number of seedlings equals the product of the number of rows and columns, with columns being 3 more than rows; learners form a quadratic equation from this context and solve it by factorisation to find the actual grid dimensions.

LESSON 1 (80 minutes (double lesson)): Anchoring Phenomenon & Forming Algebraic Expressions

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To introduce learners to the anchoring phenomenon — Achieng's chair-arrangement problem at a Kisumu prize-giving ceremony — and to build the foundational skill of using letters to represent unknown quantities and forming algebraic expressions from everyday Kenyan word statements.
Knowledge	Learners will know that: (a) unknown quantities in real-life situations can be represented using letters (variables); (b) algebraic expressions are formed by translating word statements into mathematical notation; (c) when two related unknowns have a fixed product, the relationship can be written as an equation — and that equation may involve a squared term.
Skills	Learners will be able to: (a) form algebraic expressions from different situations; (b) identify unknown quantities in real-life Kenyan contexts (seating arrangements, farm plots, market stalls); (c) write simple algebraic expressions and proto-equations from word statements; (d) post meaningful questions and wonderings on the Driving Question Board (DQB).
Attitudes	Learners will appreciate that algebra is a powerful tool for making sense of everyday Kenyan situations, and will show curiosity and openness when encountering unfamiliar mathematical structures such as squared unknowns.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	This is Lesson 1 of the sub-strand. It launches the anchoring phenomenon — Achieng's chair-arrangement problem at a Kisumu school prize-giving ceremony — and opens the Driving Question Board. Learners do not yet solve any equation; they explore why a seating problem resists simple division, represent unknown dimensions with letters, and form the algebraic expression that will become the quadratic equation in later lessons. Every subsequent lesson will return to Achieng's problem and build on the expressions formed here.
Safety Notes	No physical safety hazards. Remind learners to be respectful when sharing ideas during brainstorming; no ridicule of peers' responses. If digital devices are used to search for contexts, remind learners to observe online safety guidelines (do not share personal information; use only approved school platforms).

B. LESSON OVERVIEW

Learners are introduced to Achieng's chair-arrangement problem at a Kisumu prize-giving ceremony (120 guests; columns exceed rows by 7). Through a structured predict–observe–explain cycle, they brainstorm unknown quantities in everyday Kenyan contexts (chairs, maize-farm plots, market stalls at Kongowea Market), practise representing unknowns with letters, and translate word statements into algebraic expressions. They encounter the moment Achieng's problem resists simple arithmetic and begin to see why a squared unknown is needed. The lesson closes with the formal opening of the Driving Question Board, where learners post their questions and wonderings, anchoring the inquiry for the entire sub-strand. No quadratic equation is solved in this lesson; the goal is wonder, language, and expression-formation.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Learners listen to a read-aloud of Achieng's story (the Kisumu prize-	1. Read the phenomenon narrative aloud expressively, pausing after	Elicit-and-conflict: surfacing prior strategies (trial-and-error, division)	Observable indicator: Can the learner articulate a method (even

 VIDEO: Factorial and counting seat arrangements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	giving ceremony, 120 chairs, columns exceed rows by 7). Working individually for 3 minutes, they write on a sticky note or in their exercise books: (a) their best guess for the number of rows and columns, and (b) how they tried to find those numbers. They are not expected to be correct — the goal is to surface existing strategies and create cognitive conflict.	'...she keeps getting decimals or the wrong totals.' Ask: 'Before we do any mathematics — just using your instinct — what do you think the number of rows might be? Write it down. You have 3 minutes, working alone.' [START TIMER — 3 minutes silent individual think time.] 2. After 3 minutes, cold-call 4–5 learners (spread across the room — front, back, left, right): 'Akinyi, what was your guess and how did you try to find it?' [WAIT 10 seconds after each answer before responding or moving to the next learner.] 3. Record guesses on the chalkboard in two columns: GUESS FOR ROWS METHOD USED. Do not correct any response yet. Affirm the thinking: 'I can see some of you tried dividing 120 by different numbers — that is a very sensible first move.' 4. Pose the conflict question: 'Raise your hand if your method gave you a whole-number answer that also meant the columns were exactly 7 more than the rows.' [Pause — typically few or no hands.] 'Interesting. So the guess-and-check approach is struggling. Let us find out why — and whether algebra can rescue us.'	and creating visible cognitive conflict when those strategies fail. This motivates the need for a new algebraic tool and gives the teacher diagnostic information about learners' entry points.	an incorrect one) for finding the rows? Are learners attempting arithmetic (division of 120)? Misconceptions to note: learners who add 7 to 120, or who treat 'columns exceed rows by 7' as multiplication. Record names of learners who show no strategy — target them for support in Phase 2.
Observe Phase  VIDEO: Factorial and counting seat arrangements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos	Learners engage in two structured activities that build the observational evidence base: ACTIVITY A — 'Kenyan Contexts Gallery Walk' (15 min): Around the classroom, the teacher has posted 5 large cards, each describing a real Kenyan situation with an unknown quantity: (1) A maize farmer in the Rift Valley plants	1. Before the gallery walk: 'You are going to visit 5 situations posted around the room. At each card, with your partner, answer the three questions written at the top of the card. You have 15 minutes total — about 3 minutes per card. Go.' 2. Circulate continuously during the gallery walk. At Card 4 (Achieng's problem), stop and listen to pair conversations. If a	Representation-mapping: learners repeatedly translate between a word situation (verbal), a chosen letter (symbolic), and an algebraic product expression (mathematical). The gallery walk across 5 Kenyan contexts builds pattern recognition — the same algebraic structure appears in farming, market-vending, cooking, and seating. This prepares	Observable indicators: (a) Does the learner correctly identify the smaller unknown as the variable? (b) Can they write the second unknown as an expression in terms of the first (e.g., $r + 7$, not $r \times 7$)? (c) Do they write the product as $r(r + 7)$ rather than $r + (r + 7)$? Circulating teacher checks: scan the Product column — if learners write $r + (r+7) = 120$ instead of

 HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	seedlings in a rectangular plot — the number of columns is 5 more than the number of rows and the total is 84 seedlings; (2) A mama mboga at Kongowea Market in Mombasa arranges 60 tomatoes in a rectangular grid where columns exceed rows by 4; (3) A tailor in Kericho cuts rectangular pieces of fabric with area 48 cm^2, where the length exceeds the width by 2 cm; (4) Achieng's original problem — 120 chairs, columns exceed rows by 7; (5) A school tuck shop in Nairobi arranges 72 mandazi in a rectangular tray where columns are 1 more than twice the rows. In pairs, learners visit each card, read the situation, and write in their books: (i) What is unknown? (ii) What letter would you use to name it? (iii) Can you write the other unknown in terms of that letter? ACTIVITY B — 'Achieng's Sketch Table' (10 min): Back at their seats, learners complete a guided table (provided on a worksheet or drawn on the board): Situation Unknown chosen Expression for rows Expression for columns Product. They fill in all 5 situations from the gallery walk.	pair writes 'rows = r, columns = $r + 7$', affirm quietly: 'Hold on to that — that is going to be very important.' If a pair is stuck, prompt: 'Which quantity is smaller — rows or columns? Let that smaller one be r.' 3. After the gallery walk, direct learners back to seats: 'Let us organise what you noticed.' Cold-call 3 different pairs to share their expressions for Cards 1, 3, and 4. [WAIT 12 seconds after each share.] Record their expressions on the board exactly as stated, then ask the class: 'Does anyone want to refine or challenge what was written? [WAIT 10 seconds.]' 4. For Activity B, model the first row of the table together using Card 2 (Kongowea tomatoes): 'Unknown: let r = number of rows. Columns = $r + 4$. Product = $r(r + 4) = 60$.' Write this on the board. 'Now complete the other four rows with your partner — 10 minutes.' Circulate and check. 5. After Activity B: 'Look at the Product column. Every single product expression has the unknown appearing twice — once on its own and once inside the bracket. In our next phase, we will explore what that means.'	learners to generalise.	$r(r+7) = 120$, note this as a target misconception for Phase 3.
Explain Phase  VIDEO: Factorial and counting seat arrangements Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear	Whole-class sensemaking focused on Achieng's problem. Learners use their Activity B table as reference. Part A — Expression formation (10 min): Learners are guided to write and expand Achieng's product expression: if rows = r, then columns = $r + 7$, and the total chairs = $r(r + 7) = 120$. They expand the bracket: $r^2 + 7r = 120$,	1. 'From Activity B, let us focus on Achieng's card. Call out: what did you write as the expression for total chairs?' [Cold-call 2 learners. WAIT 12 seconds each.] Write $r(r + 7) = 120$ on the board. 2. 'Expand the left side — multiply out the bracket. What do we get?' [WAIT 10 seconds — take 1 volunteer, then cold-call 1 quiet learner.] Write $r^2 + 7r = 120$. 3. 'Now rearrange so that zero is	Bridging-and-naming: the teacher bridges from the concrete (chairs, rows, columns) to the abstract (r, $r+7$, $r^2+7r-120=0$) step by step, naming the new object (quadratic equation) only after learners have constructed it themselves. This gives ownership of the term. Part B uses 'refutation' — explicitly addressing why a familiar strategy (division) fails — to deepen understanding of why the new	Observable indicators: (a) Can the learner expand $r(r+7)$ correctly to r^2+7r? (b) Can the learner rearrange to standard form with zero on the right? (c) Can the learner replicate the process for a new context (Cards 1 and 3) with minimal prompting? Exit check: teacher scans the two equations written for Cards 1 and 3 — look for correct choice of variable, correct expression for the second

Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	and then rearrange: $r^2 + 7r - 120 = 0$. The teacher names this a quadratic equation for the first time. Part B — 'Why can't we just divide?' (8 min): In groups of 3, learners revisit the Phase 1 chalkboard list of guesses and methods. Their task: explain in one sentence why dividing 120 by a single number cannot solve Achieng's problem. Groups share out. The teacher consolidates: 'Because the two unknowns are related — one is 7 more than the other — dividing 120 by one number gives us one answer, but we need two numbers that multiply to 120 AND differ by 7. That relationship is captured by $r^2 + 7r - 120 = 0$.' Part C — Connecting back to all 5 contexts (5 min): Learners write the quadratic equation for two other situations from the gallery walk (teacher assigns Cards 1 and 3). They compare with a partner.	on the right.' [WAIT 10 seconds.] Write $r^2 + 7r - 120 = 0$. 'This is called a quadratic equation. Say it with me — quadratic equation.' [Choral response.] 'The word quadratic comes from the Latin for square, because the unknown r is squared — r^2. We will learn to solve this in the coming lessons. For now, we have written it.' 4. For Part B: 'In your groups, I want one sentence — why can dividing 120 by one number not solve Achieng's problem? Discuss for 2 minutes, then your group spokesperson will share.' [WAIT — circulate, listen.] Cold-call 2 groups. Consolidate with the explanation above. 5. For Part C: 'Write the quadratic equation for the Rift Valley maize farm (Card 1) and the Kericho fabric (Card 3). You have 5 minutes, working with your partner.' Circulate. After 5 minutes, cold-call one pair per equation to write on the board. Class checks together. 6. Transition statement: 'We now have quadratic equations from Kenyan situations. We do not yet know how to solve them — that puzzle is what will drive us for the rest of this sub-strand. Let us capture our questions.'	equation structure is necessary.	unknown, correct expansion.
Driving Question Board (DQB) Creation  VIDEO: Factorial and counting seat arrangements Source: Khan Academy (English - US curriculum) Search ARES	The class formally opens the Driving Question Board. Each learner receives two sticky notes (or tears a page into two slips). On the FIRST slip they write one question they are wondering about after today's lesson — something they do not yet know how to answer. On the SECOND slip they write one thing they now understand that they did not	1. 'We are going to open our Driving Question Board — this will be with us for every lesson in this sub-strand. It will show us how our thinking grows.' Reveal or create the DQB with the three column headings. 2. 'Take two slips. On the first, write a genuine question you are wondering about — something today's lesson raised that you	Question-driven inquiry anchoring: the DQB externalises learners' metacognitive state — what they know, what they wonder — and makes the learning trajectory visible. Grouping similar wonder questions teaches learners that mathematical inquiry is organised, not random, and that their individual questions are shared by the class community.	Observable indicators: (a) Is the learner's wonder question genuinely about the mathematics (not a procedural request like 'when is the test?')? (b) Does the 'what I now know' statement reflect the lesson's key idea (using letters for unknowns, forming expressions, naming quadratic equations)? Teacher reads all slips after class to identify learners with

for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	understand at the start of the lesson. Both slips are posted on the class DQB (a large sheet of paper or a dedicated section of the board), which is divided into three columns: WHAT WE WONDER WHAT WE NOW KNOW WHAT WE STILL NEED TO LEARN. The class does a brief 'gallery read' of the wonder questions and groups similar ones together.	cannot yet answer. On the second, write one thing you now understand. You have 3 minutes — silent, individual.' [START TIMER.] 3. After 3 minutes: 'Come up and post both slips in the correct columns.' Allow 2 minutes for posting. 4. Read aloud 5–6 wonder questions from the WHAT WE WONDER column, selecting a spread of question types. Ask the class: 'Which of these questions feel similar to each other?' Group 2–3 clusters physically on the board. Suggested clusters to anticipate and name: (i) How do we solve $r^2+7r-120=0$? (ii) What if there are two answers — which one counts? (iii) Where else in Kenya do quadratic equations appear? 5. Point to the driving question posted prominently above the DQB: 'How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?' Say: 'Every lesson in this sub-strand, we will return to this board and move questions from WONDER to KNOW. By the end, Achieng will have her seating arrangement — and you will know exactly why the equation works.' 6. Photograph the DQB (teacher's phone or school tablet) for documentation.		deep misconceptions or unexpectedly advanced thinking — use this to differentiate the next lesson.
Model Building Phase  VIDEO: Factorial and counting seat	Learners create the first version of their 'Achieng's Algebra Model' — a personal diagram they will update every lesson. In their exercise books, they draw a rectangle labelled with r rows and	1. 'In your exercise books, draw a large rectangle. This rectangle represents Achieng's seating block.' [Demonstrate on the board.] 2. 'Label the shorter side r — that is the number of rows. Label the	Cumulative model revision: the annotated rectangle is a scientific/mathematical model in the NGSS sense — a representation that captures relationships between quantities	Observable indicators: (a) Is the rectangle correctly labelled with r and $r+7$ (not r and 7, or r and $r \times 7$)? (b) Is the equation inside the rectangle written as a product $(r(r+7)=120)$ and then expanded

arrangements Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	$(r+7)$ columns, write the equation $r(r+7)=120$ inside the rectangle, and expand it to $r^2+7r-120=0$ below. Around the rectangle they annotate: (a) what r represents, (b) what $r+7$ represents, (c) what 120 represents, (d) a question mark where the solution will go. This model is the beginning of a cumulative class anchor chart that will grow each lesson as new solution methods are added.	longer side $r+7$ — that is the number of columns. Write the area — or in our case the total chairs — inside: $r(r+7)=120$.' [Model on board as learners draw.] 3. 'Below the rectangle, expand and rearrange to write the quadratic equation. Add your four annotations around the rectangle.' [Circulate for 4 minutes — check that all four annotations are present.] 4. Cold-call one learner to read their four annotations aloud. [WAIT 10 seconds before asking for a second learner to add or refine.] 5. 'This model is not finished — it has a question mark. In Lesson 2 we will learn a technique that will let us replace that question mark with actual numbers. Until then, Achieng is waiting.' [Post a class version of the model on the anchor chart beside the DQB.] 6. Assign brief home connection (not graded): 'Find one situation at home or in your neighbourhood this evening — it could involve arranging items, a piece of land, a market display — where two unknown lengths multiply to give a known total. Write it in your book exactly as we did today: choose a letter, write both expressions, write the product equation. Bring it to Lesson 2.'	and has explicit gaps (the question mark) that future learning will fill. Returning to and revising this same model every lesson makes the growth of understanding visible and tangible.	correctly below? (c) Are all four annotations present and accurate? (d) Does the question mark appear in an appropriate place? Teacher collects 5 books at random at the end of the lesson to assess model quality and identify gaps before Lesson 2.
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions:\n1. PHENOMENON UPTAKE: Did learners engage genuinely with Achieng's story, or did it feel like a contrived 'story problem'? What adjustments to the narrative would make it feel more real to learners in this specific school context?\n2. VARIABLE CHOICE: How many learners defaulted to writing columns as the variable rather than rows? Does this matter mathematically? How did you handle it?\n3. DQB QUALITY: Read all the wonder slips. Are learners asking questions about the mathematics (e.g., 'how do we find r ?', 'what if r is negative?') or surface questions (e.g., 'what is the formula?')? What does this tell you about their readiness for Lesson 2?\n4. MISCONCEPTION CHECK: Were there learners who wrote $r + (r+7) = 120$ instead of $r(r+7) = 120$? Plan a targeted intervention at the start of Lesson 2 using the Kongowea tomato context to revisit the difference between sum and product.\n5. HOME CONNECTION: In the next

lesson, create space (5 minutes) for 3–4 learners to share the real-life situations they found at home. This validates the relevance of the mathematics and enriches the class's collective knowledge of Kenyan contexts.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Achieng, a student volunteer at a Kisumu prize-giving ceremony, could not find whole-number rows and columns for 120 chairs (columns exceed rows by 7) using guess-and-check or simple division. Across 5 Kenyan contexts — from Rift Valley maize farms to Kongowea Market tomato displays to Kericho fabric pieces — the same pattern appeared: two related unknowns whose product is known.
What did I learn?	When one unknown quantity exceeds another by a fixed amount and their product is known, we can represent the smaller unknown with a letter (e.g., r), write the larger as an expression (e.g., $r + 7$), and form a product equation ($r(r + 7) = 120$). Expanding and rearranging gives a quadratic equation — $r^2 + 7r - 120 = 0$ — where the unknown appears squared. This structure cannot be solved by simple division because both unknowns are linked.
How does this explain the phenomenon?	A quadratic equation arises naturally whenever two related linear unknowns multiply to give a fixed total. The squared term appears because we multiply r by itself (indirectly, through expanding $r(r+7)$). Algebra gives us a precise language to write the relationship — turning Achieng's seating puzzle into $r^2 + 7r - 120 = 0$ — even before we know how to solve it. Solving this equation (finding the value of r) is the central challenge for the rest of the sub-strand.

LESSON 2 (80 minutes): From Expressions to Quadratic Expressions: Recognising and Writing $ax^2 + bx + c$

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners revisit the Kisumu prize-giving seating arrangement phenomenon and discover that multiplying two related linear expressions produces a squared term, enabling them to recognise, form, and write quadratic expressions in standard form $ax^2 + bx + c$ from real-life Kenyan contexts.
Knowledge	Learners know that: (i) a quadratic expression contains a squared variable as its highest power; (ii) the standard form of a quadratic expression is $ax^2 + bx + c$, where $a \neq 0$; (iii) multiplying two linear expressions of the form $(x + p)(x + q)$ always yields a quadratic expression; (iv) quadratic expressions differ from linear expressions because of the x^2 term.
Skills	Learners are able to: (i) translate a real-life rectangular arrangement or area problem into a product of two linear expressions; (ii) expand the product of two linear expressions by applying the distributive property to obtain a quadratic expression; (iii) identify the values of a , b , and c in the standard form $ax^2 + bx + c$; (iv) distinguish quadratic expressions from linear expressions by inspecting the highest power of the variable.
Attitudes	Learners appreciate that a simple real-life arrangement problem — such as seating guests at a school ceremony or planning a maize plot — can be elegantly captured by a quadratic expression, and they develop confidence in using algebraic notation to represent situations they encounter in their communities.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	In Lesson 1, learners explored the Kisumu chair-arrangement phenomenon and recognised that rows $\times$ columns = 120, with columns exceeding rows by 7, cannot be solved by simple division. In Lesson 2, learners move from that numerical puzzle to algebraic representation: they discover that letting rows = x forces columns = $x + 7$, and the product $x(x + 7) = 120$ generates a squared term. By expanding similar products from a rectangular maize plot in Eldoret and a mandazi-dough tray context, they pin down the meaning of $ax^2 + bx + c$ and update the Driving Question Board with a new, more precise question about what to do once a quadratic expression is formed.
Safety Notes	No physical hazards. Ensure learners do not share sharp pencils or geometrical instruments without care. If a projector or digital device is used, handle cables safely and avoid leaving devices unattended.

B. LESSON OVERVIEW

Learners open by briefly recalling the Kisumu seating-arrangement phenomenon from Lesson 1 (rows $\times$ columns = 120, columns exceed rows by 7). The teacher guides them to assign the variable x to the number of rows and derive that columns = $x + 7$, making the product $x(x + 7)$ the algebraic representation of 120 chairs. Through a structured Predict–Observe–Explain cycle, learners then expand $x(x + 7)$ and two further Kenyan-context products — a rectangular maize plot in Eldoret (x metres by $x + 5$ metres, area = 84 m^2) and a mandazi-dough tray $((x + 3) \text{ cm by } (x - 2) \text{ cm})$ — to discover that every such product yields a squared term. They classify the resulting expressions as quadratic, identify a , b , and c in $ax^2 + bx + c$, and contrast these with linear expressions. The lesson closes with DQB updating and a cumulative model sketch connecting today's algebraic forms to the original seating diagram.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Graphing systems of inequalities Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners are shown a recap slide of the Kisumu seating diagram from Lesson 1 (rows $\times$ columns = 120; columns = rows + 7). Working individually for 3 minutes, each learner writes in their exercise book: (a) if the number of rows is called x, what expression represents the number of columns? and (b) write the equation that says the product equals 120. They then share their prediction with a shoulder partner for 2 minutes. The class is cold-called for three different predictions. Learners are next shown a second scenario on the board: 'A farmer in Eldoret has a rectangular maize plot. The length is 5 metres more than the width. The area is 84 m^2. If the width is x metres, write an expression for the length and then write the area equation.' Learners write predictions individually for 3 minutes before a whole-class share.	1. Display the Kisumu chair diagram (drawn on the board or projected). Say: 'Before we go any further, I want each of you to work alone — no talking yet.' Allow 3 minutes of silent individual writing. 2. After 3 minutes, say: 'Now turn to your shoulder partner and compare what you wrote. You have 2 minutes.' Use the 2 minutes to circulate and note 2–3 interesting or divergent responses. 3. Cold-call exactly 3 learners (aim for 1 correct, 1 partially correct, 1 incorrect or different notation) using a random name stick. For each, ask: 'Tell us what you wrote and why you wrote it that way.' Do NOT evaluate yet — simply record each response on the board under the heading PREDICTIONS. Allow 10 seconds of WAIT TIME after each response before moving on. 4. Introduce the Eldoret maize plot scenario by reading it aloud clearly and slowly. Write on the board: 'Width = x metres. Length = _____. Area = _____ = 84.' Ask learners to complete the blanks individually in 3 minutes. 5. Cold-call 2 more learners. Record their expressions on the board next to the chair predictions. Say: 'We are going to test whether these predictions are correct as we work through today's lesson.'	Strategy: Predict-then-Record Wall. By writing predictions publicly on the board without evaluation, learners externalise their prior knowledge and create a visible reference point. This activates schema from Lesson 1 and creates cognitive dissonance when predictions differ — the dissonance motivates careful observation in the next phase. Connecting both scenarios (chairs and maize plot) immediately signals that today's pattern is not limited to one context.	Observable indicator 1: Can the learner write 'columns = $x + 7$' (or equivalent) from the verbal description? Observable indicator 2: Can the learner write $x(x + 7) = 120$ without prompting? Observable indicator 3: Can the learner write $x(x + 5) = 84$ for the Eldoret maize scenario? Teacher notes which learners write linear expressions instead of products (e.g., writing $2x + 7 = 120$) — these learners need targeted support in Phase 3.
Observe Phase  VIDEO: Graphing systems of inequalities Source: Khan Academy (English - US curriculum)  Search ARES	Learners carry out three structured expansion activities in pairs, recorded in their exercise books under the heading 'Expanding Products of Two Linear Expressions'. Activity A — Chairs (Kisumu):	1. Write all three contexts on the board (or display a prepared chart). Read each scenario aloud with expression. Say: 'With your partner, expand each product step by step. Show every step — do not skip. You have 12 minutes.' Set a visible timer. 2. Circulate actively. After 4 minutes, scan for pairs who	Strategy: Structured Parallel Cases. By expanding three different Kenyan-context products (one single-bracket, one single-bracket with different constant, one double-bracket) side by side and recording them in the same table format, learners observe the invariant pattern — every result	Observable indicator 1: Correct application of the distributive property in all three expansions (x^2 term present, like terms collected). Observable indicator 2: The three-column table completed with 'highest power = 2' in every row. Observable indicator 3: At least one member of each pair can

for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Expand $x(x + 7)$ step by step using the distributive property. Record each step: $x \times x = ?$, $x \times 7 = ?$, then write the full expression. Activity B — Maize Plot (Eldoret): Expand $x(x + 5)$. Record each step as above. Activity C — Mandazi Tray: A mandazi vendor in Kericho uses a rectangular tray of dough. She describes the tray as '$(x + 3)$ cm long and $(x - 2)$ cm wide.' Expand $(x + 3)(x - 2)$ using the distributive property: first expand $x(x - 2)$, then expand $3(x - 2)$, then add. For each result, learners fill in a three-column table in their books: Expression Expanded Form Highest Power of x After completing the table, pairs discuss: 'What is the same about all three expanded forms?'	are stuck on Activity C (two-bracket expansion). Pause the class briefly: 'I notice some of us are unsure about Activity C. Here is a reminder — I will expand the first bracket only: $(x + 3)(x - 2) = x(x - 2) + 3(x - 2)$. Now each of you finish from there.' Resume the 12-minute timer. 3. At the 12-minute mark, cold-call one pair per activity (3 pairs total) to write their expanded form on the board. Allow 10 seconds of WAIT TIME after each pair writes before asking the class: 'Does everyone agree? Show thumbs up, thumbs sideways, or thumbs down.' 4. Address any errors immediately and explicitly. If a learner writes $x^2 + 7$ instead of $x^2 + 7x$, say: 'Look carefully — when we multiply x by 7, we get $7x$, not 7. Can someone explain why?' Cold-call 2 learners. Allow 10 seconds WAIT TIME each. 5. Once all three correct expanded forms are on the board, ask: 'Fill in your three-column table now. Then with your partner, answer: what is the same about all three?' Allow 3 minutes of pair discussion.	has an x^2 term as the highest power. The table makes the comparison visual and systematic, supporting inductive generalisation without the teacher stating the pattern prematurely.	articulate 'every expansion gives an x^2 term' during the pair-discussion step. Teacher flag: any pair where Activity C still shows an error after the whole-class correction — these pairs receive a one-to-one check during Phase 3 seatwork.
Explain Phase  VIDEO: Graphing systems of inequalities Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES	Learners engage in three connected explain activities. Part A — Class Definition Build (8 min): Using the three expanded forms on the board, the teacher guides learners to name the general structure $ax^2 + bx + c$. Learners copy the definition into their books and then identify a, b, and c for each of the three expressions, recording in a second table: Expression a b c	1. Part A: Point to the three expanded forms on the board. Say: 'Look at all three. What letter should we use for the number in front of x^2? In front of x? For the number with no x?' Build the label $ax^2 + bx + c$ live on the board using learner suggestions. Cold-call 3 learners to state the values of a, b, c for the chairs expression, the maize plot expression, and the mandazi tray expression respectively. Allow 10 seconds WAIT TIME after each question. Say explicitly: 'Note — if there is no x term, then $b = 0$. Let us	Strategy: Concept Mapping through Contrasting Cases. By sorting six expressions into LINEAR vs QUADRATIC after seeing the three expanded forms, learners construct the boundary of the concept 'quadratic expression' through contrast rather than mere definition. This deepens conceptual understanding beyond rote recognition. The two new Kenya-context formation tasks then require learners to apply the concept productively, consolidating the define–contrast–apply cycle.	Observable indicator 1: All six expressions correctly sorted with written justification referencing the x^2 term. Observable indicator 2: Values of a, b, c correctly identified for all three Phase 2 expressions. Observable indicator 3: Context 1 gives $x(x + 4) = x^2 + 4x$; Context 2 gives $(2x + 1)(x - 3) = 2x^2 - 5x - 3$ — both in standard form with correct a, b, c. Teacher flag: any learner who writes $b = 4$ instead of $b = 4$ for Context 1 but cannot explain why — indicates procedural without conceptual understanding; revisit the

for similar readings	Part B — Distinguishing Quadratic from Linear (7 min): Learners are given six expressions on a worksheet (or written on the board): (i) $3x + 5$ (ii) $x^2 + 4x - 3$ (iii) $2x$ (iv) $x^2 - 9$ (v) $x + 7$ (vi) $2x^2 - x + 1$ They sort these into two groups — LINEAR and QUADRATIC — and write a one-sentence justification for each group. Part C — New Context Formation (10 min): Working individually, learners form and expand quadratic expressions for two new Kenyan contexts written on the board: Context 1: 'Omondi is arranging githeri tins on a shelf. The number of tins in each row is 4 more than the number of rows. If there are x rows, write the quadratic expression for the total number of tins.' Context 2: 'A school garden near Lake Victoria has a rectangular flower bed. The length is $(2x + 1)$ metres and the width is $(x - 3)$ metres. Expand and write the expression in standard form.' Learners write their working in full in their books.	check: does that happen in any of our three expressions?' Pause for 15 seconds of WAIT TIME. Cold-call 1 learner. 2. Part B: Read the six expressions aloud. Say: 'Sort them — you have 5 minutes, working alone.' After 5 minutes, cold-call 2 learners to share their sorting. Ask the class: 'Who sorted differently? Raise your hand.' Address any disagreement. Ask: 'What is the ONE feature that makes an expression quadratic and not linear?' Cold-call 3 learners. Write the consensus answer on the board: 'A quadratic expression has x^2 as its highest power term ($a \neq 0$).' 3. Part C: Read both contexts aloud slowly. Say: 'Work alone. Show every step of your expansion. You have 8 minutes.' Circulate — target learners you flagged in Phase 2. After 8 minutes, cold-call 1 learner for each context to write their answer on the board. Ask the class to check using thumbs. Correct any errors. Say: 'Notice — we are now writing quadratic expressions from situations right here in Kenya. Achieng in Kisumu did exactly this with the chairs. What did we actually do to get from the problem to the expression?' Allow 10 seconds WAIT TIME. Cold-call 2 learners.		distributive property with that learner.
Driving Question Board (DQB) Creation  VIDEO: Graphing systems of inequalities Source: Khan Academy (English - US curriculum)  Search ARES	Each learner receives two sticky notes (or tears a sheet into two pieces). On the first, they write one thing they NOW understand about the Kisumu chair problem that they did not understand in Lesson 1. On the second, they write one NEW question that has emerged — something they still do not know. Learners post their notes on the class DQB, which is organised	1. Distribute or direct learners to tear paper. Say: 'Two notes each. You have 3 minutes — write your own ideas, not your partner's.' Enforce quiet individual writing. 2. Direct learners to post notes on the DQB. Allow 2 minutes for posting. 3. Stand at the DQB and read 4–5 'wonder' notes aloud verbatim, without attribution. After each, ask: 'Does anyone else have	Strategy: DQB Clustering. Making thinking public on the DQB externalises metacognition and creates a shared class narrative about the progress of understanding. Clustering 'wonder' notes shows learners that their confusion is collective, not individual, and creates a purposeful roadmap for upcoming lessons — directly advancing the	Observable indicator 1: Each learner produces at least one substantive 'know' statement that references $ax^2 + bx + c$ or the expansion process. Observable indicator 2: 'Wonder' questions are algebraically sensible (e.g., asking about solving, about negative values of x, about which solution fits the context) rather than off-topic. Observable indicator 3: The

for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	in two columns: WHAT WE NOW KNOW and WHAT WE STILL WONDER. After posting, the teacher reads 4–5 of the 'wonder' notes aloud and groups them into emerging themes (e.g., 'How do we solve $x^2 + 7x = 120$?' and 'Can x have two values?'). These grouped wonders are circled and labelled as targets for future lessons.	a similar question? Raise your hand.' This validates the questions as shared rather than individual. 4. Use a marker to circle clusters of similar questions. Label the biggest cluster — likely around solving the equation — as 'Target: Lesson 3.' Say: 'Today we moved from a number puzzle to an algebraic expression. Our DQB shows us we still need to find out how to get a value of x from that expression. That is exactly where Achieng and Mr Otieno are heading.' 5. Leave the DQB visible for the rest of the lesson.	driving question storyline.	teacher can read at least 3 distinct conceptual themes from the posted notes — indicating a spread of genuine engagement rather than superficial copying.
Model Building Phase  VIDEO: Graphing systems of inequalities Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners open their 'Running Model' page (started in Lesson 1). In Lesson 1 they drew the Kisumu seating grid with rows and columns labelled numerically. Today, they add an algebraic layer to that same model: they label the rows as x and the columns as $x + 7$, write the product $x(x + 7)$ next to the grid, expand it to $x^2 + 7x$, and draw a double-headed arrow from the grid to the expression with the label 'expanding gives a quadratic expression.' They then add a small second sketch — the Eldoret maize plot rectangle, labelled x and $x + 5$, with $x^2 + 5x$ written beneath it and the note 'same pattern.' At the bottom of the model page they write the standard form $ax^2 + bx + c$ and annotate a, b, c on their Kisumu expression.	1. Say: 'Open your Running Model page. You drew the chair grid in Lesson 1. We are now going to add algebra to that same picture.' Give a 30-second pause for learners to find their model page. 2. Model the annotation LIVE on the board by drawing the grid quickly, labelling rows as x and columns as $x + 7$, and writing $x(x + 7) = x^2 + 7x$ with a double-headed arrow. Say: 'Copy this onto your own model page — but use your own handwriting and add colour if you can. You have 5 minutes.' 3. After 5 minutes, ask: 'Now add the Eldoret maize plot sketch. Two minutes.' Circulate and check that the model pages show both the visual (grid/rectangle) and the algebraic expression. 4. In the final 2 minutes, ask 2 learners to hold up their model pages and describe one thing they added today. Cold-call 1 learner who has not spoken yet this lesson. 5. Close by connecting to the driving question: 'Achieng now has the equation $x^2 + 7x = 120$ written on her paper. She has gone from a seating puzzle to an algebraic expression.	Strategy: Cumulative Visual Model Revision. Returning to the same model page across lessons makes learning progression visible to the learner. Adding an algebraic layer on top of the Lesson 1 numerical/visual layer externalises the conceptual shift from arithmetic to algebra and reinforces that the expression emerged from the original real-life context — not from an abstract rule. The second sketch (Eldoret) generalises the pattern across contexts.	Observable indicator 1: The model page shows both the grid and the algebraic expression $x(x + 7) = x^2 + 7x$ with correct expansion. Observable indicator 2: The Eldoret rectangle sketch is present with $x(x + 5) = x^2 + 5x$. Observable indicator 3: a, b, c are correctly annotated on at least one expression. Teacher collects 5 model pages at random at the end of class for a quick review before Lesson 3.

		What does she need to do next to find x ? Hold that question — it is exactly what we tackle in Lesson 3.'	
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During Phase 2, did learners expand Activity C — the double-bracket mandazi tray product — correctly without scaffolding, or did the mid-activity pause become necessary? If most pairs needed the pause, consider whether a brief worked example of double-bracket expansion should be added to the opening of Lesson 3. (2) In Phase 3 Part B (sorting linear vs quadratic), how many learners initially classified $x^2 - 9$ as 'not quadratic' because there is no x term? This is a key misconception to address — b can equal 0. If more than one-third of the class made this error, open Lesson 3 with a 3-minute revisit. (3) Review the 5 collected model pages: do learners' algebraic annotations match the visual context, or are they writing expressions disconnected from the grid? Disconnected annotations suggest the link between the real-life rectangle and the algebraic product has not been internalised. (4) Count the number of distinct conceptual themes on the DQB 'wonder' column. If learners are only asking 'how do we solve it?' that is a good sign — they have understood today's content. If learners are still asking 'why does multiplying give x^2 ?' then the Phase 2 observation activity needs an additional concrete example at the start of Lesson 3. (5) Note any learner who was cold-called but could not respond even after 10 seconds of wait time — arrange brief individual support before the next lesson, focusing on the distributive property with numerical examples (e.g., $3 \times (4 + 2) = 3 \times 4 + 3 \times 2$) before moving to algebraic ones.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	When we multiplied the number of rows (x) by the number of columns ($x + 7$) for the Kisumu chair arrangement, and when we multiplied the sides of the Eldoret maize plot (x by $x + 5$) and the Kericho mandazi tray ($(x + 3)$ by $(x - 2)$), every product produced an expression with an x^2 term as the highest power.
What did I learn?	Multiplying two linear expressions always produces a quadratic expression of the form $ax^2 + bx + c$, where $a \neq 0$. The x^2 term appears because $x \times x = x^2$. We can identify a (coefficient of x^2), b (coefficient of x), and c (the constant term) in any expanded quadratic expression. A quadratic expression differs from a linear expression because its highest power of x is 2, not 1.
How does this explain the phenomenon?	The reason the Kisumu seating problem — and any rectangular-arrangement or area problem where one dimension exceeds the other by a fixed number — naturally produces a quadratic expression is that area (or total count) equals length $\times$ width, and when both dimensions depend on the same unknown x , their product must contain x^2 . This is why Achieng's teacher Mr Otieno said the relationship involves a squared unknown: it is unavoidable once the two dimensions are expressed in terms of the same variable.

LESSON 3 (80 minutes): Forming Quadratic Equations from Real-Life Situations

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To enable learners to read, interpret, and translate real-life Kenyan scenarios into correctly formed quadratic equations, and to identify the defining features of a quadratic equation.
Knowledge	Learners will know that a quadratic equation is formed when a real-life relationship between two unknown quantities involves a product that yields a squared term, and that it takes the standard form $ax^2 + bx + c = 0$ where $a \neq 0$.
Skills	Learners will be able to: (a) define unknown quantities using a single variable; (b) express the relationship between two related unknowns algebraically; (c) form a quadratic equation from a word problem involving areas, seating arrangements, or seedling grids; (d) verify their equation by checking it matches a given numerical condition.
Attitudes	Learners will appreciate that writing a precise mathematical equation is more powerful and reliable than guess-and-check, and will demonstrate respect for peers' reasoning by listening actively during pair and group discussions.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	In Lesson 1 learners explored Achieng's seating puzzle and saw why guess-and-check fails. In Lesson 2 they built algebraic expressions for unknown dimensions. Now in Lesson 3 they take the critical next step: translating three distinct Kenyan real-life scenarios — a Rift Valley fencing problem, the Kisumu prize-giving seating arrangement, and a Kericho seedling grid — into fully formed quadratic equations. By the end of the lesson, learners can explain why the equation for Achieng's chairs must have a squared term, setting up Lessons 4–5 where they will learn to solve such equations by factorisation.
Safety Notes	No physical hazards. Remind learners to observe online safety guidelines if digital devices are used to search for additional real-life applications of quadratic equations (KICD PCIs: Online Safety).

B. LESSON OVERVIEW

Learners work through three carefully sequenced Kenyan context cards — (1) a rectangular fencing scenario on a Rift Valley farm, (2) Achieng's 120-chair prize-giving layout at the Kisumu school, and (3) a seedling grid at a Kericho tea nursery — to practise forming quadratic equations. They predict the algebraic structure before reading, form equations in pairs, compare with another pair, and present to the class. The teacher surfaces and addresses common misconceptions (treating the equation as linear, placing the squared term on both sides, forgetting to set the equation equal to zero). A whole-class discussion consolidates the three defining features of a quadratic equation. Learners update the Driving Question Board (DQB) to flag that they now know HOW to write the equation but still need to know HOW TO SOLVE it — bridging to Lessons 4 and 5. The lesson closes with a cumulative model revision in which learners annotate their seating-layout sketch with the equation they have now formed.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase	Each learner receives a half-page 'Scenario Starter' card showing	Distribute Scenario Starter cards face-down. Say: 'Do NOT turn the	Strategy: Elicit-Before-Explain. By committing predictions to writing	Observable indicator: Learner has written two distinct algebraic

 VIDEO: Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	ONLY the numerical facts for Scenario A: 'A farmer in the Rift Valley has 40 metres of fencing wire. She wants to enclose a rectangular plot so that the length is 4 metres more than the width. What is the area of the plot?' Learners individually write (a) what they think the two unknown quantities are, (b) how they would express the length in terms of the width, and (c) whether they think multiplying the two expressions will give a linear or a non-linear result — and why. They record predictions in their exercise books before any class discussion.	card over until I say so.' After 30 seconds of settling, say: 'Turn your card over. Read it once silently. Now, in your book, write your three predictions. You have 3 minutes — go.' Circulate and scan predictions without correcting. After 3 minutes, cold-call 3 learners: 'Wanjiku, what did you choose as your unknown?' — 'Kamau, did you predict linear or non-linear — tell us your reason in one sentence.' — 'Atieno, same question, do you agree or disagree with Kamau?' Allow 10–12 seconds WAIT TIME after each question before accepting an answer. Do NOT reveal the correct answer yet. Record the split of opinions (linear vs. non-linear) on the board as a tally.	before instruction, learners activate prior knowledge from Lessons 1–2 (expressions for related unknowns) and create a personal stake in finding out whether their prediction is correct. The visible tally of class opinions creates productive cognitive conflict that the Observe phase will resolve.	expressions (e.g. 'width = w, length = w + 4') before instruction begins. If the majority of learners write only a single number or leave the algebraic expression blank, pause and ask: 'In Lesson 2, how did we write Achieng's columns in terms of her rows? Use the same idea here.' This indicates readiness to proceed to the Observe phase.
Observe Phase  VIDEO: Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners work through all three scenario context cards in a structured pair-then-group sequence. Each card is released one at a time. Card A (Rift Valley fencing — perimeter constraint): width = w, length = w + 4, perimeter = 40 m → form the equation. Card B (Achieng's chairs — Kisumu prize-giving): rows = r, columns = r + 7, total chairs = 120 → form the equation. Card C (Kericho seedling grid — tea nursery): seedlings per row = s, seedlings per column = s – 3, total seedlings = 180 → form the equation. For each card, pairs first write the equation independently (2 min), then compare with the adjacent pair and resolve differences (2 min), then one spokesperson shares the equation with the class. Learners record all three equations in a three-column table: Scenario Let statement Equation formed.	Release Card A. Say: 'Step 1 — write your Let statement: let the width equal what? Step 2 — write the length in terms of your unknown. Step 3 — use the perimeter formula. Step 4 — expand and rearrange so one side is zero. You and your partner have 2 minutes.' After 2 minutes: 'Now turn to the pair behind you — compare your equations. If they differ, decide together which is correct and be ready to explain why. 90 seconds.' Cold-call one group: 'Omondi, read your equation aloud exactly as you wrote it.' Write the equation on the board verbatim, even if incorrect. Ask the class: 'Do we agree? Thumbs up, thumbs sideways, or thumbs down — show me now.' Address any thumbs-down. Explicitly annotate on the board: circle the squared term and label it 'degree 2 term'; circle the linear term; circle the constant. Repeat the release-pair-group-share cycle	Strategy: Three-Card Parallel Case Analysis. Presenting three structurally similar but contextually different scenarios (perimeter/area, seating product, seedling product) allows learners to abstract the common mathematical structure — the quadratic form — across contexts. Writing equations in a comparison table and presenting them side-by-side makes the pattern visible and supports inductive reasoning toward a generalisation.	Observable indicators: (1) Learner's Let statement uses a single variable and correctly expresses the second quantity in terms of it. (2) Learner's equation has a squared term on one side and a constant on the other before rearranging. (3) After rearranging, the equation is set equal to zero. Flag any learner who writes two different variables (e.g. 'let width = w, let length = l') without a linking expression — prompt: 'How does l relate to w in this problem? Can you write l using only w?'

		for Cards B and C. After Card C, say: 'Look at all three equations on the board. What do they ALL have in common? Discuss with your partner for 30 seconds.' WAIT 30 seconds. Cold-call 2 learners. Establish the three defining features on the board: (1) exactly one squared term, (2) the coefficient of the squared term is not zero, (3) the equation is set equal to zero.		
Explain Phase  VIDEO: Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners tackle two consolidation tasks individually, then discuss in groups of four. Task 1 (Misconception Sort): Learners are given four 'student work' strips showing four different attempted equations for Card B (Achieng's chairs). One is correct; three contain typical errors — (i) equation is linear ($r + r + 7 = 120$), (ii) squared term appears on both sides ($r^2 + 7r = r^2 + 120$), (iii) equation is not set equal to zero ($r^2 + 7r = 120$ left unsimplified as 'the answer'). Learners identify which is correct, name the error in each incorrect strip, and write a one-sentence correction. Task 2 (Extend to a new context): 'A mandazi seller in Mombasa arranges mandazi in a rectangular tray. The number of mandazi in each row is 5 more than the number of rows. There are 84 mandazi in total. Form a quadratic equation.' Learners work individually (3 min) then compare with a partner.	Distribute the Misconception Sort strips. Say: 'You have 4 minutes — individually sort these strips: one correct, three wrong. For each wrong one, write the name of the error in the margin.' After 4 minutes: 'Share in your group of four. If you disagree, use the three features on the board to arbitrate.' After 2 minutes of group discussion, cold-call one group: 'Njeri, which strip did your group mark as correct, and what was wrong with Strip i?' Write the error names on the board: 'Error 1: Linear equation — forgot to multiply, just added. Error 2: Squared term cancels — equation collapses. Error 3: Not set equal to zero — incomplete form.' Explicitly connect each error back to the phenomenon: 'If Achieng used Strip i to find her rows, what number would she get? Would it work for 120 chairs? Why not?' Allow 10 seconds WAIT TIME. Transition to Task 2. After 3 minutes of individual work, cold-call 2 different learners to write their equation on the board. Ask the class: 'Are both equations equivalent? How can we check?' Guide learners to verify by substituting the positive root ($r = 7$, so rows = 7, columns = 12,	Strategy: Misconception-Driven Explanation (Identify–Name–Correct). Presenting worked errors rather than only correct examples forces learners to articulate WHY a step is wrong, which deepens conceptual understanding more than simply solving correctly. Naming errors ('linear error', 'cancellation error', 'incomplete form error') gives learners a shared vocabulary for self-monitoring in future tasks.	Observable indicators: (1) Learner correctly identifies the one valid equation among the four strips without teacher prompting. (2) Learner's written error label matches the mathematical flaw (not just 'it looks different'). (3) For Task 2, learner's equation is in the form $n^2 + 5n - 84 = 0$ or equivalent. If fewer than half the class correctly identifies the Misconception Sort in 4 minutes, pause the class and work through Strip i together before releasing Task 2.

		product = 84).		
Driving Question Board (DQB) Creation  VIDEO: Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Each learner receives two sticky notes (or writes in a DQB section of their book if physical notes are unavailable). On the FIRST sticky note (green/positive): they write one thing they can now do that they could NOT do at the start of Lesson 1 — specifically about forming equations. On the SECOND sticky note (orange/question): they write the most important question that is STILL unanswered for them — the class expectation is that most questions will be about HOW TO SOLVE the equation now that they know how to form it. Learners post or share their notes. The teacher clusters them on the DQB under two headers: 'WE CAN NOW...' and 'WE STILL NEED TO KNOW...'. The teacher explicitly maps the 'still need to know' cluster to the lesson sequence: 'Lesson 4 — factorising; Lesson 5 — solving by factorisation; Lessons 6–7 — applying solutions.'	Say: 'Take 2 minutes — write your two sticky notes. Green: one thing you can now do. Orange: your biggest remaining question.' After 2 minutes, invite 4–5 volunteers to read their orange notes aloud before posting. As they read, the teacher groups similar questions on the DQB visually. Say: 'Look at the orange cluster — almost everyone is asking some version of: HOW DO WE FIND THE VALUE OF r (or w or s)? That is exactly what Achieng is asking in our phenomenon. We now know how to WRITE her problem as an equation. Starting next lesson, we learn how to CRACK the equation open.' Point to the DQB headline question: 'How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?' Say: 'Today we conquered the WRITING part. The SOLVING part starts in Lesson 4.'	Strategy: Public Knowledge Tracking (DQB Update). Making the boundary between known and unknown knowledge visible on a shared board externalises metacognition and gives learners a sense of narrative progress through the unit. Connecting the 'still need to know' questions explicitly to the upcoming lesson sequence builds anticipation and purpose for Lessons 4–5.	Observable indicators: (1) Learner's green note describes a specific mathematical action ('I can write a quadratic equation using one variable') rather than a vague statement ('I understand more'). (2) Learner's orange note frames a mathematical question about solving or factorising — not a procedural complaint. If most orange notes are blank or off-topic, prompt: 'If I give you the equation $r^2 + 7r - 120 = 0$ right now, could you find r? What would you need to know to do that?'
Model Building Phase  VIDEO: Proof: Rhombus diagonals are perpendicular bisectors Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX)	Learners return to the cumulative model they have been building since Lesson 1 — Achieng's hand-drawn rectangular seating layout sketch. In this lesson's revision, they add three annotations directly onto or beside the sketch: (A) The Let statement: 'Let the number of rows = r; number of columns = $r + 7$.' (B) The product equation: '$r(r + 7) = 120$.' (C) The standard quadratic form: '$r^2 + 7r - 120 = 0$.' Below the sketch, learners write a one-sentence 'Model Status Note': 'I can now write the equation that describes Achieng's problem. I cannot yet find the value of r — I	Say: 'Open to your cumulative model — Achieng's seating sketch. Add the three annotations I am writing on the board now.' Write on the board: '(A) Let statement — (B) Product equation — (C) Standard form.' Give learners 5 minutes of silent individual work. Circulate and check: (i) that the standard form has the squared term first and equals zero, (ii) that the Let statement uses r (not two separate variables), (iii) that the Model Status Note is honest — learners should NOT claim they can solve yet. After 5 minutes, cold-call 2 learners to read their Model Status	Strategy: Cumulative Annotated Model Revision. Requiring learners to add specific, lesson-generated knowledge to a persistent visual model (the seating sketch) reinforces that each lesson advances a single coherent investigation rather than being an isolated topic. The 'Model Status Note' develops epistemic honesty — learners distinguish what they know from what they do not yet know, a key scientific and mathematical practice.	Observable indicators: (1) All three annotations are present and mathematically correct on the sketch. (2) Standard form annotation equals zero (not just '$r^2 + 7r = 120$'). (3) Model Status Note explicitly states that solving is still needed. Collect 5–6 books at random to check model annotations before Lesson 4 as a pre-assessment of readiness for factorisation.

Source: CK-12 Search ARES for similar readings	need a method to solve $r^2 + 7r - 120 = 0$.' Learners who finish early annotate their model with a second scenario of their choice (fencing or seedlings) to show the equation formed.	Note aloud. Affirm: 'This is exactly right — a scientist or mathematician does not pretend to know more than they do. Achieng does not yet know r . Neither do we. But we will.' Close the lesson by directing learners to the DQB: 'Our model is growing. Each lesson we add one more layer. Next lesson — factorisation — we will finally start to crack $r^2 + 7r - 120 = 0$ open.'		
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D. TEACHER REFLECTION

After the lesson, consider the following questions: (1) In the Misconception Sort (Phase 3), which of the three errors was most common among your learners — the linear error, the cancellation error, or the incomplete-form error? What does this tell you about the algebraic prerequisite knowledge to revisit before Lesson 4? (2) Were learners able to write a correct Let statement using a single variable, or did many revert to two separate variables? If the latter, plan a 5-minute warm-up at the start of Lesson 4 using a simple numerical example (e.g. 'One number is 3 more than another — write both using one variable') before introducing factorisation. (3) Review the orange DQB sticky notes: are learners' questions about solving focused on factorisation (as expected) or are there questions about what a 'solution' means? If the latter, plan to spend extra time in Lesson 5 on interpreting both roots in context. (4) Did the three-scenario parallel structure (fencing, chairs, seedlings) give sufficient variety, or did learners treat all three as identical? If they seemed mechanical, consider opening Lesson 4 with a novel scenario (e.g. a githeri bean-planting grid) to test genuine transfer before teaching factorisation.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners read three Kenyan real-life scenario cards (Rift Valley fencing, Kisumu prize-giving seating, Kericho seedling grid) and recorded the numerical relationships described in each — including the total area or total count and the way the two dimensions were related.
What did I learn?	Learners learned that when two related unknown quantities are multiplied together to give a known total, the resulting equation always contains a squared term (degree 2), making it a quadratic equation. They identified three defining features of a quadratic equation: exactly one squared term, a non-zero coefficient for that term, and the equation set equal to zero. They also named three common formation errors: the linear error, the cancellation error, and the incomplete-form error.
How does this explain the phenomenon?	Learners explained why Achieng's seating problem — and the other two scenarios — cannot be described by a simple linear equation: because the two dimensions are both expressed in terms of the same unknown, their product inevitably produces a squared term. They annotated their cumulative seating-layout model with the Let statement, the product equation, and the standard quadratic form $r^2 + 7r - 120 = 0$, and wrote a Model Status Note acknowledging that forming the equation is now within reach but solving it requires a new method.

LESSON 4 (80 minutes (2 × 40-minute lessons)): Factorisation of Quadratic Expressions: The Product-and-Sum Method

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	To explicitly teach learners the product-and-sum strategy for factorising quadratic expressions derived from real-life contexts, and to verify factorisations by expanding.
Knowledge	Learners will know that a quadratic expression of the form $x^2 + bx + c$ can be factorised by finding two numbers whose product equals c and whose sum equals b , and that factorisation reverses expansion.
Skills	Learners will be able to: (i) identify the product and sum values from a quadratic expression; (ii) apply the product-and-sum method to factorise quadratic expressions; (iii) verify a factorisation by expanding the brackets; (iv) form and factorise quadratic expressions from real-life contexts such as seedling arrangements and seating layouts.
Attitudes	Learners will appreciate that a systematic algebraic method (factorisation) is more reliable than guess-and-check when solving real-world arrangement and dimension problems, and will develop confidence in collaborative mathematical reasoning.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	In Lessons 1–3, learners formed quadratic expressions from Achieng's chair-arrangement problem and explored how multiplying brackets produces quadratic expressions. In this lesson, learners work backwards — they are explicitly taught the product-and-sum method to factorise quadratic expressions (e.g. $n^2 + 3n - 108$ from a seedling context, and $n^2 + 7n - 120$ from the Kisumu ceremony). This is the critical factorisation tool they will use in Lesson 5 to solve the equation and finally answer Achieng's question.
Safety Notes	No physical safety hazards. Remind learners of respectful peer-discussion norms. If digital devices are used to verify expansions, remind learners to follow school online-safety guidelines.

B. LESSON OVERVIEW

Learners are explicitly taught the product-and-sum strategy for factorising quadratic expressions. Starting from the Kisumu prize-giving phenomenon ($n^2 + 7n - 120$) and a seedling-arrangement context ($n^2 + 3n - 108$), they identify the required product and sum, list factor pairs, and write the factorised form. Peer discussion and collaborative practice on a range of expressions reinforce the method. Learners verify every factorisation by expanding the brackets. The DQB is updated with the emerging question: "Can factorisation always work?" This lesson directly advances the driving question by giving learners the algebraic tool needed to break apart and solve the quadratic equation that describes Achieng's seating problem.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Concentration Source: PhET	Learners are shown two expressions on the board derived from previous lessons: (i) $n^2 + 7n - 120$ (Achieng's chair layout) and (ii) $n^2 + 3n - 108$ (a maize seedling)	Write both expressions clearly on the board in two colours. Say verbatim: 'We have seen these expressions before — they came from Achieng's chairs and from a	Prediction Activation (Eliciting Prior Knowledge): By asking learners to sketch bracket forms before instruction, the teacher surfaces intuitive understanding of	Observable indicator: Every learner has a written prediction and at least one attempted bracket sketch in their exercise book. Teacher circulates during the 3-

Interactive Simulations (English) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	arrangement at a farm in Kericho where total seedlings = 108, columns exceed rows by 3). They are asked: 'Without calculating, predict — do you think these expressions can be broken into two simpler bracket expressions? Write your prediction in your exercise book, and sketch what two brackets might look like.' Learners work silently for 3 minutes, writing their intuitive guesses (e.g. $(n + ?)(n - ?)$). A show-of-hands poll is taken: 'How many predict both expressions can be factorised? How many predict only one? How many predict neither?' Tally marks are recorded on the board to revisit at the end of the lesson.	maize farm in Kericho. Today we are going to learn a powerful method to break them apart. Before I teach you anything, I want your honest prediction.' Allow 3 minutes of silent individual thinking — do NOT prompt further. After 3 minutes, conduct the show-of-hands poll and record tallies visibly. Cold-call 3 learners (one from each prediction group) using name cards: 'Otieno, tell us one reason for your prediction.' Allow 10–12 seconds WAIT TIME after each cold-call before accepting responses. Do not affirm or correct at this stage — say: 'Interesting — let us keep these predictions and test them today.'	factorisation from earlier expansion work in Lessons 2–3. Recording class predictions publicly creates cognitive commitment — learners are motivated to confirm or revise their thinking, building metacognitive awareness.	minute silent period and notes (without correcting) whether learners are attempting $(n + a)(n + b)$ forms, whether they recognise a negative constant requires one positive and one negative bracket, and whether any learner is already correctly guessing the factor pairs. This informs pacing during the Explain phase.
Observe Phase VIDEO: Video: Concentration Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	Learners receive a printed or ARES-screen activity card titled 'Seedling Rows in Kericho.' It reads: 'A farmer in Kericho planted maize seedlings in a rectangular arrangement. The total number of seedlings is 108. The number of columns is 3 more than the number of rows. Let the number of rows = n. (a) Write an expression for the number of columns. (b) Write an equation for the total number of seedlings. (c) Expand and simplify to form a quadratic expression.' Learners work in pairs for 6 minutes to form the expression $n^2 + 3n - 108$. They then examine a pre-prepared factor-pair table (provided on the activity card) listing all integer factor pairs of 108: (1,108), (2,54), (3,36), (4,27), (6,18), (9,12), and their negatives. Learners are asked to OBSERVE: 'For each pair, calculate the sum. Record the sum in the third column of the table.' They complete the table and	Distribute activity cards (or display on ARES). Say verbatim: 'Work with your partner. You have 6 minutes to form the quadratic expression — use what we learned in Lessons 2 and 3.' Set a visible timer. After 6 minutes, cold-call one pair to share their expression — allow 10 seconds WAIT TIME. Confirm: 'Yes — $n^2 + 3n - 108$. Now look at the factor-pair table. Your job is purely to OBSERVE — fill in the sum column. Do not try to factorise yet.' Circulate actively, crouching to pair level. Look for: pairs who correctly identify -9 and $+12$ as the target pair (product = -108, sum = $+3$). After 5 minutes of table-completion, say: 'Raise your hand if you found a pair whose sum is exactly $+3$.' Cold-call 2 pairs to share. Allow 12 seconds WAIT TIME. Record the identified pair ($-9, +12$) on the board in a box labelled 'KEY PAIR.'	Structured Data Collection (Factor-Pair Table): Providing the pre-built factor-pair table removes arithmetic burden and focuses learner attention on the PATTERN — that one specific pair satisfies both the product condition (-108) and the sum condition ($+3$). This makes the logic of the product-and-sum method visible before the algorithm is named, building inductive understanding rather than rote rule-following.	Observable indicator: The completed factor-pair table with the sum column filled in and the pair ($-9, +12$) circled. Teacher checks 4–5 books during circulation. If fewer than 60% of pairs correctly identify the key pair, the teacher pauses and models the sum calculation for the first two rows before allowing pairs to continue — a responsive mid-activity scaffold.

	circle the pair whose sum equals +3.			
Explain Phase VIDEO: Video: Concentration Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	The teacher formally names and models the product-and-sum method using the seedling expression $n^2 + 3n - 108$. Learners follow in their exercise books, writing each step as it is modelled. Step 1 — Identify a, b, c: coefficient of n^2 is 1, $b = +3$, $c = -108$. Step 2 — State the target: 'Find two numbers whose product = $c = -108$ AND sum = $b = +3$.' Step 3 — Use the observed key pair: -9 and +12. Step 4 — Rewrite: $n^2 - 9n + 12n - 108$. Step 5 — Group and factor: $n(n - 9) + 12(n - 9)$. Step 6 — Final factorised form: $(n + 12)(n - 9)$. Step 7 — VERIFY by expanding: $(n + 12)(n - 9) = n^2 - 9n + 12n - 108 = n^2 + 3n - 108$. ✓ Learners then apply the method independently to Achieng's expression $n^2 + 7n - 120$, following the same seven steps. After independent work (8 minutes), pairs compare answers and reconcile differences. The class then works through two further practice expressions set in Kenyan contexts: (i) $x^2 + 5x - 84$ (number of mandazi trays arranged by a caterer in Mombasa) and (ii) $x^2 - 2x - 63$ (plot dimensions in a housing estate in Nakuru). For each, learners must verify by expanding.	Model the seedling expression SLOWLY on the board, writing each step with a label. Say verbatim at Step 2: 'This is the heart of the method — we need TWO numbers. They must MULTIPLY to give us c, and ADD to give us b. Both conditions must be true at the same time.' Pause for 12 seconds. 'Write that in a box in your notes — product = c, sum = b.' After modelling, say: 'Now you try Achieng's expression — $n^2 + 7n - 120$ — on your own. 8 minutes. Silent, individual work.' Set timer. During circulation, do NOT give answers — if a learner is stuck at Step 2, ask: 'What is the product you are looking for? What is the sum?' After 8 minutes, say: 'Compare with your partner for 2 minutes.' Cold-call 3 learners to share their factorised forms and verification — allow 12 seconds WAIT TIME each time. For the Mombasa and Nakuru practice expressions, use Think-Pair-Share: 3 min individual → 2 min pair → cold-call 2 pairs per expression. After all four expressions, return to the board tally from Phase 1 and ask: 'Were your predictions correct?' Pause 10 seconds. Cold-call 2 learners who predicted incorrectly: 'What surprised you?'	Gradual Release with Verification Loop (I Do → You Do → We Check): The teacher models once with the seedling expression, then releases to independent practice with the phenomenon expression (Achieng's chairs), then to peer-supported practice with new contexts. Crucially, the VERIFY step (expanding to check) is built into every round — this creates a self-correction habit and connects factorisation back to the expansion knowledge from Lessons 2–3, reinforcing the inverse relationship between the two operations. Returning to the Phase 1 predictions at the end closes the inquiry loop and strengthens metacognitive reflection.	Observable indicators: (1) Learner exercise books show all 7 steps for each expression, including the verification expansion. (2) During cold-calls, learners can state the product and sum targets before listing factor pairs — not just the final answer. (3) Exit-check: teacher asks 3 randomly selected learners to state verbally 'What are the two conditions a factor pair must satisfy?' If any learner cannot articulate both conditions, the teacher re-models Step 2 for the whole class using a brief second example before moving to Phase 4.
Driving Question Board (DQB) Creation VIDEO: Video: Concentration Source: PhET	Learners revisit the class Driving Question Board displayed at the front of the room. In pairs, they discuss: 'What have we learned today that helps us answer the driving question — how can writing and solving a quadratic equation help us find unknown dimensions	Direct learners to the DQB. Say verbatim: 'In the last three lessons we built the quadratic equation. Today we learned how to break it apart. Update the board — what do we now know, and what are we still wondering?' Allow 4 minutes for pair discussion and card	Public Knowledge Tracking (DQB Update Routine): Separating 'what we now know' (green) from 'what we still wonder' (orange) makes the progression of understanding visible to every learner in the room. The emergent question 'Can factorisation always work?' is not	Observable indicator: Every pair produces both a green UPDATE card and an orange QUESTION card. Teacher scans green cards to confirm that at least 80% correctly reference the product-and-sum method and/or a correctly factorised expression from today's

Interactive Simulations (English) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	and arrangements?' Each pair writes one UPDATE card (green sticky note or ARES digital card) summarising new understanding, e.g. 'We can factorise $n^2 + 7n - 120$ into $(n + 15)(n - 8)$ using the product-and-sum method.' They also write one NEW QUESTION card (orange sticky note), prompted by: 'What are you still wondering? Can factorisation always work? What if no whole-number factor pair exists?' New question cards are placed on the DQB under the heading 'Still Wondering.' The class reads three new question cards aloud. The teacher highlights 'Can factorisation always work?' as the key open question to be explored in Lesson 5.	writing. Collect cards and place them on the board yourself, reading each aloud. After placing all cards, cold-call 2 learners whose 'Still Wondering' cards raise the question of whether factorisation always works: 'Can you say more about what made you write that?' Allow 10 seconds WAIT TIME. Close this phase by saying: 'This question — can factorisation always work — is exactly what Achieng and Mr Otieno will face in Lesson 5. Keep it in mind.'	answered yet — intentionally preserving productive tension that drives engagement in Lesson 5. This routine also develops learner agency by positioning students as co-constructors of the inquiry trajectory.	lesson. If update cards show misconceptions (e.g. 'we just split the middle term randomly'), these are addressed at the start of Lesson 5.
Model Building Phase VIDEO: Video: Concentration Source: PhET Interactive Simulations (English) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	Learners return to the cumulative 'Seating Arrangement Model' they began in Lesson 1 — a diagram or annotated sketch showing Achieng's problem evolving into algebraic form. In this lesson they add a new layer: the factorised form. Individually, learners update their model to show: (i) the quadratic expression $n^2 + 7n - 120$ labelled on the rectangle (from Lesson 3), (ii) the product-and-sum working (product = -120, sum = $+7$; factor pair: -8 and $+15$), (iii) the factorised form $(n + 15)(n - 8)$ written inside or beside the rectangle, (iv) the verification expansion confirming correctness. They annotate the model with the sentence: 'The expression $(n + 15)(n - 8)$ tells us the two bracket factors, but we have not yet set the equation equal to zero and solved it — that is Lesson 5.' Learners share their updated models with a partner and give one piece of	Say verbatim: 'Take out your cumulative seating model. We are adding the factorisation layer today. You have 7 minutes to update it — include the product-and-sum working, the factorised form, and the verification.' Set timer. Circulate and look for: (1) correct factor pair identification, (2) correct factorised form, (3) verification expansion present. After 7 minutes, direct peer feedback using the sentence stem written on the board. Cold-call one pair to share their model annotation under a visualiser or by holding it up. Ask the class: 'Does this model now show us everything we need to find how many rows and columns Achieng needs?' After 10 seconds WAIT TIME, guide learners to articulate: 'We have the factorised form but we still need to set it equal to zero and solve — the model is not yet complete.' This creates explicit	Cumulative Model Revision (Layered Representation Building): By physically adding the factorisation layer to the same model built across Lessons 1–4, learners see the problem-solving journey as a coherent sequence rather than isolated skills. The annotation 'not yet equal to zero' preserves the incompleteness of the model honestly, making the next lesson's goal (solving the equation) feel necessary and motivated rather than arbitrary.	Observable indicator: Updated models contain all four components — labelled expression, product-and-sum working, factorised form, and verification. Peer feedback using the sentence stem is audible and specific. Teacher collects 5 models at random to review before Lesson 5 — if factorised forms are incorrect or verification steps are missing, a 5-minute correction activity will open Lesson 5 before new content is introduced.

	feedback using the stem: 'I can see that you understand ____ because ____.'	anticipation for Lesson 5.		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) During the Observe phase, were learners able to complete the factor-pair sum column independently, or did most pairs need scaffolding? If scaffolding was frequently needed, consider providing a partially completed table in Lesson 5's practice set. (2) During the Explain phase, how many learners could articulate BOTH conditions (product = c AND sum = b) during the formative exit-check? If fewer than 70% could do so, open Lesson 5 with a 5-minute retrieval practice activity before introducing equation solving. (3) Were the DQB 'Still Wondering' cards substantive? Did learners raise the question of what happens when no integer factor pair exists (non-factorisable expressions)? If not, the teacher should plant this question explicitly at the start of Lesson 5 to prepare learners for the quadratic formula later in the sequence. (4) Did the Kenyan contexts (Kericho seedlings, Mombasa mandazi, Nakuru housing) feel authentic to learners or forced? Note any learner comments about context for future lesson design. (5) Was the 80-minute block sufficient, or did the independent practice in Phase 3 require more time? If time was short, the Nakuru practice expression can be set as homework with a verification requirement.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed a factor-pair table for the expression $n^2 + 3n - 108$ (Kericho maize seedling arrangement), completing the sum column for all integer factor pairs of 108 and identifying the pair $(-9, +12)$ whose product is -108 and sum is $+3$.
What did I learn?	Learners learned the product-and-sum method for factorising quadratic expressions: identify two numbers whose product equals the constant term and whose sum equals the coefficient of the linear term, rewrite the middle term using these two numbers, group and factorise, then verify by expanding. They applied this method to Achieng's chair-arrangement expression $n^2 + 7n - 120 = (n + 15)(n - 8)$.
How does this explain the phenomenon?	Learners explained that factorisation reverses expansion and that every factorisation can be verified by multiplying the brackets back out. They updated the DQB with the open question 'Can factorisation always work?', and annotated their cumulative seating-arrangement model with the factorised form and product-and-sum working, noting that the equation must still be set equal to zero and solved — the goal of Lesson 5.

LESSON 5 (80 minutes): Solving Quadratic Equations by Factorisation: Finding and Interpreting Both Roots

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners apply the factorisation method to solve quadratic equations, set each factor equal to zero to find both roots, and critically interpret which root(s) make sense in the context of real-life problems such as Achieng's chair arrangement and a farm-fencing scenario.
Knowledge	Learners know that a quadratic equation set equal to zero can be factorised into two linear factors; that setting each factor equal to zero yields two potential solutions (roots); and that the Zero Product Property justifies this procedure.
Skills	Learners can rearrange a quadratic equation into standard form, factorise it correctly, apply the Zero Product Property to find both roots, and evaluate each root against the constraints of the real-life context (e.g., rejecting negative dimensions).
Attitudes	Learners appreciate that mathematics is a reasoning tool — finding two answers is not an error but a feature, and choosing the contextually valid answer requires critical thinking and respect for real-world constraints.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	In Lesson 5 of the evidence-gathering sequence, learners return to Achieng's prize-giving seating problem and the farm-fencing scenario established in earlier lessons. Having learned to form and factorise quadratic expressions (Lessons 1–4), they now complete the solution process: they set the equation equal to zero, factorise, apply the Zero Product Property, obtain both roots, and decide which root is meaningful for the chair layout. This advances the driving question by showing that solving the equation fully — including interpreting the answers — is what allows Achieng and Mr Otieno to plan the actual seating arrangement.
Safety Notes	No physical hazards. Remind learners to respect peers during whole-class discussion; no ridicule of incorrect answers. If digital devices are used to verify solutions, observe school ICT-use guidelines and online safety protocols.

B. LESSON OVERVIEW

Learners work through the complete factorisation-solution process for quadratic equations arising from real-life Kenyan contexts. Starting with Achieng's chair-arrangement equation ($x^2 + 7x - 120 = 0$) and a rectangular farm-fencing problem from Kericho, learners rearrange equations to standard form, factorise into two linear factors, apply the Zero Product Property to obtain both roots, and then engage in a teacher-facilitated sensemaking discussion to decide which root(s) are valid. The lesson uses a Predict–Solve–Interpret–Discuss cycle. Pair work, cold-calling, mini-whiteboards (or exercise books held up), and a class summary table anchor collaborative sensemaking throughout. By the end, every learner can articulate not just the algebraic procedure but also why one root may be rejected in context — a key threshold concept for the sub-strand.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Small sample hypothesis test	Learners are shown the equation $x^2 + 7x - 120 = 0$ on the board (Achieng's chair equation derived in Lesson 4) alongside the prompt:	Display the equation prominently: 'Here is the equation Achieng wrote for the 120-chair arrangement: $x^2 + 7x - 120 = 0$.	Prediction with public commitment (fingers poll + recorded reasoning). This activates prior knowledge about equations and creates	Teacher scans exercise books during the 3-minute individual time; looks for evidence that learners are connecting to prior knowledge

Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	'Without solving, predict — how many values of x do you think will satisfy this equation? Write your prediction and one sentence of reasoning in your exercise book.' Learners work individually for 3 minutes, then share with a shoulder partner for 2 minutes. A 'show-of-fingers' poll (1 finger = one solution, 2 fingers = two solutions, 3 fingers = more than two) is conducted with eyes closed to reduce social pressure. Predictions are recorded on a sticky-note wall or a section of the board labelled 'Our Predictions' so they can be revisited after solving.	Before we touch algebra — predict: how many values of x satisfy this?' Allow 3 minutes of silent individual thinking (enforce quiet with a visible timer). After partner sharing, say: 'Eyes closed. Show me with your fingers how many solutions you predict.' Count responses aloud: 'I see mostly twos — interesting. Let's keep that in mind.' Cold-call 3 learners with different predictions: 'Otieno, you said one — tell us your thinking.' (WAIT TIME: 12 seconds.) 'Wanjiru, you said two — what made you say that?' (WAIT TIME: 10 seconds.) 'Kipchoge, you said more than two — explain.' (WAIT TIME: 12 seconds.) Record each reasoning statement verbatim on the board under 'Our Predictions' without evaluating. Say: 'We will test all three ideas today.'	cognitive dissonance — learners who predict 'one solution' will be productively surprised, deepening encoding of the Zero Product Property when the lesson resolves the tension.	(e.g., referencing linear equations having one solution, or recalling that a squared term 'does something different'). Finger poll gives an instant class distribution. If more than 60% predict 'one solution', teacher notes this and plans to address it explicitly during the Explain phase.
Observe Phase VIDEO: Small sample hypothesis test Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	Learners work through two structured evidence-gathering tasks in pairs, recording every step in their exercise books. TASK A — Achieng's Chair Equation: Pairs receive (or copy from the board) the equation $x^2 + 7x - 120 = 0$ with a step scaffold: Step 1 — Confirm standard form ($ax^2 + bx + c = 0$). Identify a, b, c. Step 2 — Find two numbers whose product = -120 and sum = $+7$. Step 3 — Rewrite the middle term and factorise by grouping. Step 4 — Write the factorised form: $(x + 15)(x - 8) = 0$. Step 5 — Apply the Zero Product Property: state 'If $A \times B = 0$, then $A = 0$ or $B = 0$.'	Distribute or display the step scaffold for Task A. Say: 'Work in your pairs. I want every single step written — do not skip. You have 12 minutes.' Circulate actively. After 4 minutes, pause the class: 'Raise your hand if you have found two numbers whose product is -120 and sum is $+7$. Good — keep going.' (This is a temperature check, not a cold-call.) After 12 minutes, cold-call 2 different pairs to write their Step 6 on the board simultaneously — one on the left half, one on the right half — without talking. Say: 'Everyone compare your Step 6 with what you see on both sides. Are they the same? Different? Why?' (WAIT TIME: 10 seconds.) Address any discrepancy immediately. Transition to Task B: 'Now let us	Worked-example pair practice with a visible step scaffold (cognitive apprenticeship). The two-task sequence moves from a given equation (Task A) to a problem requiring equation formation (Task B), deliberately building transferability. Simultaneous board work by two pairs externalises thinking and creates a natural comparison opportunity.	Observable indicators: (1) In Task A, Step 5 — can the learner write the Zero Product Property in their own words before applying it? (2) In Task B — does the learner define a variable before writing the equation? (3) Do learners record BOTH roots even when one is negative? Red flag: learners who stop after finding $x = 8$ in Task A without finding $x = -15$. This is the primary misconception the Explain phase must address.

	Step 6 — Solve each factor: $x + 15 = 0 \rightarrow x = -15$; $x - 8 = 0 \rightarrow x = 8$. Step 7 — Write both solutions and circle them. TASK B — Farm-Fencing Problem from Kericho: 'A farmer in Kericho wants to fence a rectangular plot. The length exceeds the width by 5 metres, and the area of the plot is 84 m^2. Find the dimensions of the plot.' Pairs must: (i) define a variable, (ii) form the quadratic equation, (iii) rearrange to standard form, (iv) solve by factorisation, (v) write both roots, (vi) note both roots on their work. Both tasks completed individually in pairs; exercise books remain open on the desk.	apply the same process to a farm in Kericho. This time I am not giving you the equation — you must form it first.' Read the problem aloud twice. Say: 'Before you write any algebra, tell your partner what the unknown is.' (WAIT TIME: 10 seconds.) Cold-call 1 pair: 'Auma and Mwangi — what is your unknown?' Confirm or redirect. Allow 10 minutes for Task B. Circulate and note pairs who are struggling to form the equation versus those struggling to factorise — these are different misconceptions requiring different interventions.		
Explain Phase  VIDEO: Small sample hypothesis test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	The class transitions to a whole-group sensemaking discussion facilitated by the teacher. Three anchor questions are displayed on the board: (Q1) 'We got $x = 8$ and $x = -15$ for Achieng's chairs. Can BOTH be the number of rows? Which do we use? Why?' (Q2) 'For the Kericho farm, what were your two roots? Can both be lengths of a field? Why or why not?' (Q3) 'Mathematically, both roots are valid solutions to the equation. Why does context force us to reject one?' Learners first write a response to all three questions individually (3 minutes), then the class discusses. A volunteer scribes the class's	Open the discussion: 'We found $x = 8$ and $x = -15$ from Achieng's equation. Mathematically, both satisfy the equation. Let us check: if $x = -15$, the number of rows is -15. Wanjiru, is it possible to have -15 rows of chairs?' (WAIT TIME: 12 seconds, expect laughter/agreement.) 'So we reject $x = -15$ because...?' Cold-call 3 learners to complete the sentence: 'We reject $x = -15$ because...'. (Cycle through 3 cold-calls with 10-second wait each.) For Q2 (Kericho farm), say: 'What were the two roots your pair found for the farm problem? Kimani — read both roots.' (WAIT TIME: 10 seconds.) Display or write the correct roots ($x = 12$ and $x = -7$). Ask: 'Can a field have a width of -7 metres? Naliaka, explain.'	Structured Whole-Class Discussion anchored by a Context Check Table (a public record). The three-question sequence escalates from concrete (specific problems) to abstract (why context constrains mathematics), pushing learners toward generalisation. The table externalises reasoning and provides a reference artefact for the DQB update.	Observable indicators: (1) Can learners articulate the reason for rejection in a complete sentence without prompting? (2) Do learners spontaneously apply the same rejection logic to the farm problem, or does it need to be re-taught? (3) Listen for language: learners should use words like 'negative dimension is impossible', 'length cannot be negative', or 'rows must be a whole positive number'. If the class struggles with Q3, teacher provides a bridging example: 'If I told you $2 \times 3 = 6$, does the number 6 know whether we are counting mandazi or counting students? No — the interpreter does.'

	agreed answers on the board in a 'Context Check Table' with columns: Equation Root 1 Root 2 Rejected Root Reason for Rejection Accepted Answer. Learners copy the completed table into their exercise books. The teacher then connects back to the phenomenon: Achieng now has her answer — 8 rows and 15 columns — and Mr Otieno can confirm the layout. The class counts: $8 \times 15 = 120$. ✓	(WAIT TIME: 12 seconds.) For Q3, pose to the whole class: 'Here is the big idea — the algebra gave us two valid mathematical answers, but only one valid real-life answer. Why does the mathematics not automatically know which is right?' (WAIT TIME: 15 seconds — extended wait, this is a deep question.) Cold-call 2 learners. Summarise: 'The equation does not know we are talking about chairs or farms. WE must interpret the answer using context. This is what mathematicians — and farmers and event organisers — do.' Guide the volunteer scribe to complete the Context Check Table. Say: 'Copy this table. It is your key sensemaking record for today.'		
Driving Question Board (DQB) Creation  VIDEO: Small sample hypothesis test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Each learner writes two sticky notes (or two sentences in their exercise book if sticky notes are unavailable): – GREEN note: 'Something I can now answer about the driving question — How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations?' – ORANGE note: 'A new question OR a remaining confusion I still have.' Learners post their notes on the class DQB (a designated section of the wall or board). The teacher reads 3–4 green notes and 3–4 orange notes aloud to the class.	Say: 'Take two minutes. Write your green note: one thing you can now explain about how quadratic equations help in real life — be specific, use Achieng's chairs or the Kericho farm.' (WAIT TIME: 2 minutes, enforce silence.) 'Now write your orange note: one question you still have, or one thing that still confuses you.' (1 minute.) As learners post notes, teacher does a quick scan. Read 3 green notes aloud: 'Listen to what your classmates can now explain.' Read 3 orange notes: 'These are honest questions — the class will answer them in Lessons 6–8.' Point to any orange note that asks about 'what if you cannot	Public knowledge-tracking via the DQB (a Next Generation Science Education / inquiry learning practice adapted for mathematics). Colour-coding green (can explain) versus orange (still wondering) makes metacognition visible and communal, reinforcing that 'not yet knowing' is a legitimate and valued learning state.	Teacher reads all posted notes before the next lesson. Green notes that are vague ('I learned about equations') are flagged — those learners need a targeted follow-up question in Lesson 6. Green notes that reference specific numbers (e.g., '8 rows $\times$ 15 columns = 120') show deep contextual understanding. Orange notes revealing confusion about WHY we set the equation to zero (not just HOW) signal a concept gap to address in Lesson 6's opening.

	Orange notes that are answered by today's lesson are moved to the 'Answered' column. New orange notes are added to the 'Still Wondering' column. Learners record in their exercise books: 'DQB Update — Lesson 5: [one thing we can now answer] [one thing still open].'	factorise?' and say: 'This is a beautiful question — it is exactly what Lesson 6 will address.' Do NOT answer it now — preserve the productive tension. Say: 'The DQB shows our thinking growing. In Lesson 1, we had almost no green notes. Today we have [count them]. This is evidence of your learning.'		
Model Building Phase  VIDEO: Small sample hypothesis test Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners retrieve the cumulative 'Equation Model' diagram they have been building since Lesson 1 (a hand-drawn flowchart or concept map in their exercise books). Today's addition: they add a new box or layer labelled 'Step 4 — Solve by Factorisation' with the following elements: – Zero Product Property statement (in their own words) – The two-branch arrow showing 'Factor 1 = 0 → Root 1' and 'Factor 2 = 0 → Root 2' – A 'Context Filter' box connected by an arrow, with the label: 'Reject root if it violates real-life constraints (e.g., negative length/number)' – Achieng's chair answer circled: $x = 8$ rows, 15 columns Below the model, learners write a 2-sentence 'Model Claim': 'My model now shows that solving a quadratic equation by factorisation always gives two roots. I must check both roots against the real-life situation to decide which one(s) to accept.' In the final 3 minutes, pairs share their models and check each	Say: 'Open your exercise books to your Equation Model. We are adding the most important layer yet.' Display a completed version of the new layer on the board (teacher-prepared). Walk through each element: 'The Zero Product Property — write it in your own words, not mine.' (WAIT TIME: 90 seconds for writing.) 'The two-branch arrow — draw it. Label Root 1 and Root 2.' 'The Context Filter box — this is new today. It did not exist in Lessons 1–4 because we had not reached this step. Draw it. Write inside it: Reject if impossible in context.' After 8 minutes of individual model-building, say: 'Share with your partner. Check: does their model have the Context Filter? If not, help them add it.' (2 minutes of pair checking.) Close with a cold-call to 2 learners: 'Read me your Model Claim — the two sentences at the bottom.' (WAIT TIME: 10 seconds each.) Affirm or sharpen language. Final statement: 'Achieng can now tell Mr Otieno: 8 rows and 15 columns. The mathematics worked — and so did your thinking.'	Cumulative Model Revision (adapted from NGSS Modelling Practice, SEP-2). The model is a living artefact that learners have been building across all 8 lessons; adding the Context Filter box today makes the interpretive step of mathematics visible as a permanent, named component of the problem-solving process — not an afterthought.	Observable indicators: (1) Is the Context Filter box present and labelled in every learner's model? (2) Does the Model Claim include BOTH the algebraic outcome (two roots) AND the interpretive step (context filter)? (3) During pair-sharing, do learners correct each other's omissions — showing peer-to-peer mathematical agency? Collect 4–5 exercise books at random at the end of the lesson to review model diagrams for quality and completeness before Lesson 6.

	other's 'Context Filter' box for completeness.			
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions:

1. ZERO PRODUCT PROPERTY UNDERSTANDING: Could learners articulate WHY we set each factor to zero (the Zero Product Property) in their own words, or were they only procedurally executing the steps? If the 'why' was weak, plan a 5-minute re-explanation at the start of Lesson 6 using a numerical analogy (e.g., 'If I tell you that two numbers multiply to give zero, what must be true about at least one of them?').
2. CONTEXT REJECTION LOGIC: During the Explain phase, how readily did learners transfer the rejection logic from Achieng's chair problem to the Kericho farm problem without teacher prompting? If transfer was slow, consider adding a third example early in Lesson 6 where BOTH roots are positive and valid — to show that rejection is context-dependent, not a rule that 'negative always = reject'.
3. DQB ORANGE NOTES: What proportion of orange notes asked about cases where factorisation is difficult or impossible? This will calibrate how much scaffolding Lesson 6 (completing the square or quadratic formula) will need.
4. EQUITY OBSERVATION: Were the same 4–5 learners dominating cold-call responses? Review your cold-call records and deliberately target different learners in Lesson 6, especially learners who were silent today.
5. PHENOMENON CONNECTION: Did learners spontaneously use Achieng's name and numbers (8 rows, 15 columns, 120 chairs) in their DQB notes and Model Claims, or did they write in purely abstract algebraic language? Strong phenomenon connection is a sign of deep contextual understanding; abstract language only may indicate surface procedural learning.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	We observed that the equation $x^2 + 7x - 120 = 0$ (Achieng's chair problem) and the Kericho farm equation both produced TWO solutions when solved by factorisation.
What did I learn?	We learned that the Zero Product Property lets us set each factor equal to zero to find both roots, and that we must use the real-life context to decide which root(s) are valid — rejecting negative values where dimensions or counts cannot be negative.
How does this explain the phenomenon?	We explained why Achieng's answer is 8 rows and 15 columns (not –15 rows): the algebra gives two mathematically correct roots, but only the positive root makes sense for a physical seating arrangement. This is the 'Context Filter' step in our growing model.

LESSON 6 (40 minutes): Verification and Deeper Factorisation Practice — Checking Solutions and Contextual Validity

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners practise verifying solutions by substitution and apply factorisation to a range of quadratic equations drawn from Kenyan real-life contexts, determining when one, both, or neither solution is valid.
Knowledge	– A solution to a quadratic equation can be verified by substituting it back into the original equation and confirming the result equals zero. – A quadratic equation formed from a real-life situation may yield two mathematical roots, but the context determines how many roots are physically meaningful. – Quadratic equations arise in diverse Kenyan contexts — matatu profit, market trading, and athletics projectile motion — all sharing the same algebraic structure $ax^2 + bx + c = 0$.
Skills	– Substitute candidate solutions into the original quadratic equation and evaluate whether each side balances. – Analyse the real-life context of a quadratic equation to accept, reject, or qualify each root on grounds of feasibility (e.g., non-negative length, positive quantity). – Factorise quadratic expressions of increasing complexity using the product-and-sum method and apply the Zero Product Property to find both roots.
Attitudes	– Appreciate that mathematics demands precision — a correct algebraic answer that is contextually impossible must still be rejected. – Show intellectual honesty by acknowledging errors in peer work and correcting them constructively.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	Having formed equations (Lessons 1–3), factorised (Lesson 4), and found both roots (Lesson 5), learners now stress-test every solution through substitution and contextual reasoning, fully resolving Achieng's original puzzle about which chair-arrangement values are valid and why.
Safety Notes	No specific hazards. Ensure peer-checking discussions remain respectful; remind learners that critiquing a solution is not criticising the person.

B. LESSON OVERVIEW

This lesson is the verification and consolidation engine of the quadratic equations sequence. In Lessons 4 and 5 learners discovered how to factorise and find two roots; now they confront the disciplined follow-up question: are those roots actually correct, and do they make sense in the situation that produced the equation? The lesson opens by returning to Achieng's chair-arrangement problem from the anchoring phenomenon — learners substitute both roots ($r = 8$ and $r = -15$) into the original equation $r^2 + 7r - 120 = 0$ to confirm algebraic correctness, then argue from context why $r = -15$ must be discarded. This substitution ritual becomes the gold standard students apply to every subsequent problem.

The Observe Phase presents three progressively complex Kenyan-context problems: (1) a matatu operator in Nairobi whose weekly profit P (in hundreds of shillings) satisfies $n^2 - 3n - 40 = 0$, where n is the number of extra trips; (2) a market trader in Kisumu whose rectangular display table has area 28 m^2 and whose length exceeds width by 3 m ; and (3) a Kenyan athletics coach at Iten who models a shot-put throw with $h^2 - 5h + 4 = 0$ (in simplified time units). In each case learners factorise, find both roots, verify by substitution, and decide contextual validity. Peer-checking pairs swap workbooks and audit each other's substitution steps, practising error analysis.

The Explain Phase surfaces the generalisation: the number of valid solutions depends on the sign constraints, physical constraints, and meaning of the variable in the problem. Learners update the Driving Question Board, finally answering the sticky note from Lesson 1 — "When you get two answers, which one do you use?" — and revise their cumulative model to show the complete solution pathway from phenomenon to verified, contextually justified answer.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Video: Proportion Playground Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners are shown Achieng's two roots from Lesson 5 written on the board: $r = 8$ and $r = -15$. Before any calculation, each learner writes a one-sentence prediction in their exercise book: 'I think both/only one/neither root will satisfy the original equation when I substitute it back, because...' They then share predictions with a shoulder partner for 60 seconds.	Display on the board: 'Achieng solved $r^2 + 7r - 120 = 0$ and got $r = 8$ OR $r = -15$. Before we check: write your prediction — will substituting $r = 8$ make the equation true? Will substituting $r = -15$ make it true? Give a reason.' Allow 90 seconds of silent individual writing. Say: 'Turn to your partner — share your prediction in one sentence. You have 60 seconds.' Call on 3 cold-call students (spread across rows): ask each 'What did you predict and why?' [WAIT TIME: 10 seconds after each student speaks before responding.] Do NOT confirm or deny — respond with 'Interesting reasoning — let us test it.'	Prediction with Justification: by committing to a written prediction before computing, learners activate prior knowledge about what 'solving' means and surface any misconception that only positive roots are 'real' solutions algebraically. The subsequent dissonance (both roots satisfy the equation algebraically) creates productive cognitive conflict that the Observe Phase resolves.	Scan written predictions as learners write. Look for: (a) learners who predict only $r = 8$ is valid — these learners conflate algebraic and contextual validity (key misconception to address in Explain Phase); (b) learners who predict neither will work — these learners may not yet trust the factorisation from Lesson 5; (c) learners who correctly predict both will satisfy the algebra. Note 2–3 names from each category for targeted cold-calling later.
Observe Phase  VIDEO: Video: Proportion Playground Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar	Working in pairs, learners complete a structured Verification and Practice Worksheet with four problems of increasing complexity. Problem 1 (Achieng's chairs): substitute $r = 8$ and $r = -15$ into $r^2 + 7r - 120 = 0$, show all arithmetic, state whether each substitution balances, then decide which value works for the seating arrangement. Problems 2–4 introduce new Kenyan contexts — a Nairobi matatu operator, a Kisumu market trader, and a Iten athletics coach — each requiring full factorisation, both roots, verification by substitution, and a written	Distribute the worksheet. Say: 'Problem 1 is Achieng's problem — you already have the roots. Your job is to prove algebraically that both roots satisfy the equation, then decide contextually. Show every substitution step — no skipping.' [WAIT TIME: allow 4 minutes for Problem 1 before intervening.] Circulate and look for learners who write ' $8^2 + 7(8) = 120$ ' without subtracting 120 — prompt: 'The equation is set equal to zero — what must your substitution also equal?' For Problem 2 (matatu), read aloud: 'Mwangi operates a matatu on the Nairobi–Thika route.	Graduated Complexity with Peer Audit: the four problems are sequenced from familiar (Achieng, known roots) to novel contexts, ensuring learners first experience success with verification before encountering new factorisation challenges. The peer-audit swap externalises the checking process — learners must read another person's substitution steps sequentially, which forces attention to mathematical logic rather than answer-matching. Error analysis builds metacognitive awareness of common mistakes (e.g., forgetting the ' $= 0$ ' requirement, arithmetic	During circulation, check specifically: (1) Does the learner write the full substitution (e.g., $(8)^2 + 7(8) - 120 = 64 + 56 - 120 = 0$ ✓) or just state 'it works'? Full working is required. (2) For Problem 3, does the learner correctly form the equation $w(w + 3) = 28 \rightarrow w^2 + 3w - 28 = 0$ before factorising? Learners who jump directly to trial values without forming the equation need re-direction to Lesson 3 skills. (3) For Problem 4, do learners recognise that $h = 1$ and $h = 4$ are both non-negative and therefore both contextually possible (representing

readings	contextual verdict. After completing each problem, pairs swap books and peer-check the substitution arithmetic step-by-step, marking any error with a circled 'E' and a correction note.	His weekly profit model gives $n^2 - 3n - 40 = 0$, where n is the number of extra trips above his minimum. Factorise, find both roots, verify each by substitution, and state which value of n makes business sense.' For Problem 3 (Kisumu market trader), read aloud: 'Atieno sells vegetables at Kibuye Market. Her rectangular display table has an area of 28 m^2. The length is 3 m more than the width. Write a quadratic equation, factorise, solve, verify both roots by substitution, and state the actual dimensions.' For Problem 4 (Iten athletics), read aloud: 'Coach Kiplagat at Iten models a shot-put arc with the equation $h^2 - 5h + 4 = 0$, where h is height in metres at key instants. Factorise, solve, verify, and interpret both solutions.' After Problem 2, call on a pair to demonstrate substitution on the board [WAIT TIME: 12 seconds before accepting a volunteer]. After Problem 4, direct pairs to swap and peer-check — say: 'Circle any step where the arithmetic is wrong. Write a correction note — do not erase their work.'	sign errors in substitution).	two heights at which the shot-put is at the same elevation)? This is the lesson's richest contextual discussion point. Collect 3–4 worksheets at the end for written feedback.
Explain Phase  VIDEO: Video: Proportion Playground Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12	The class reconvenes for a whole-class sensemaking discussion. A selected pair presents their solution and contextual verdict for Problem 3 (Kisumu market trader) on the board. A second pair presents Problem 4 (Iten athletics). The class compares verdicts across all four problems and, guided by teacher questions, constructs a shared 'Validity Decision Rule' in their notes: a three-row table with columns 'Context Constraint', 'Root 1 Decision', 'Root 2 Decision'.	Ask a pre-selected pair to present Problem 3 on the board. After they finish, ask the class: 'Did your pair get the same roots? Same contextual verdict? Show of hands — who rejected the negative root?' [WAIT TIME: 10 seconds.] Then ask: 'For Problem 4 with Coach Kiplagat, how many of you kept BOTH roots as valid? Why?' Cold-call 3 students [WAIT TIME: 12 seconds each]. After discussion, say: 'Let us build a rule together. What question do you ask yourself after finding both roots?' Guide	Collaborative Sense-Making with Anchor Return: presenting peer solutions (rather than teacher solutions) keeps learners as the authority on the mathematics, while the teacher's role is to draw out the generalisation. Explicitly returning to the Lesson 1 sticky-note question closes a narrative loop in the storyline — learners experience the satisfaction of answering a question they raised themselves seven lessons ago. The three-question Validity Decision Rule provides a	Listen for: (a) Can learners articulate WHY $r = -15$ is algebraically valid but contextually invalid — not merely that 'you can't have negative rows', but that the variable r was defined as a count of rows, which is always a positive whole number? (b) Do learners spontaneously apply the three-question rule to Problem 4 and correctly justify keeping both roots? (c) Are written answers to the sticky-note question complete (mentioning both algebraic verification AND contextual

 Search ARES for similar readings	Learners then revisit the original Lesson 1 sticky-note question — 'When you get two answers, which one do you use?' — and write a full answer in their own words.	learners to articulate: (1) Does the variable represent something that must be non-negative? (2) Does the variable represent a whole number only? (3) Does the value make the physical situation impossible? Write the three-question checklist on the board as learners copy it. Then return to the phenomenon: 'Achieng had $r = 8$ and $r = -15$. Walk me through all three questions for $r = -15$.' Cold-call Achieng's scenario specifically — ask: 'Can you have -15 rows of chairs at a prize-giving ceremony in Kisumu? Explain.' [WAIT TIME: 10 seconds.] Close by saying: 'Open your books to Lesson 1 — find the sticky-note question you wrote then. Now write a complete answer to it using today's three-question rule.'	transferable heuristic that works across all subsequent quadratic contexts.	checking) or only partial? Collect written sticky-note answers as an exit indicator.
Driving Question Board (DQB) Creation  VIDEO: Video: Proportion Playground Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners return to the class Driving Question Board. They identify sticky notes placed in Lessons 1–3 that this lesson has now answered — particularly 'When you get two answers, which one do you use?' and 'Can both answers be right?' They move those notes to the 'ANSWERED' section and write the date and a brief summary answer on the back. New sticky notes added today capture any remaining uncertainties, for example: 'What if both roots are negative — does the problem have no real solution?' or 'Can a quadratic ever give more than two answers?'	Direct learners' attention to the DQB. Say: 'Look at the questions we posted in earlier lessons. Which ones can we now answer fully — not just partially?' Allow 2 minutes for learners to re-read all posted questions silently. Then invite volunteers to come to the board and physically move a sticky note to the ANSWERED column, reading the question aloud and giving a one-sentence answer. [WAIT TIME: allow up to 4 learners to do this.] Then ask: 'Are there any NEW questions today's lesson has raised for you?' Allow 2 minutes for learners to write new sticky notes and post them. Read 2–3 new notes aloud, affirming them: 'These are excellent questions — we will address some in Lesson 7 and beyond.' Photograph the updated DQB for the class record.	Driving Question Board Update (Iterative Knowledge Tracking): the physical act of moving a sticky note from 'OPEN' to 'ANSWERED' gives learners a tangible record of intellectual progress — they can see the evidence-gathering sequence accumulating understanding. New questions posted today signal that verification has opened new mathematical territory (e.g., the possibility of no real solutions), maintaining forward momentum in the storyline.	Observe which sticky notes learners choose to move and how they phrase their summary answers. A learner who can correctly summarise 'Both roots satisfy the equation algebraically, but the context determines which are valid' demonstrates full lesson mastery. A learner who moves only the 'negative root' note but not the 'can both be right?' note may still hold a partial understanding — follow up individually.

Model Building Phase  VIDEO: Video: Proportion Playground Source: PhET Interactive Simulations (English)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners open their cumulative Solution Pathway Model — the diagram they have been building since Lesson 1 showing how to move from a real-life problem to a verified, contextually justified answer. Today they add the final two steps to the model: Step 5 'Verify each root by substitution into the original equation' and Step 6 'Apply the Validity Decision Rule — check context constraints for each root'. They also annotate the model with a worked mini-example from today: the Kisumu market trader problem ($w^2 + 3w - 28 = 0$), showing both roots, both substitution checks, and the contextual verdict. Learners who finish early add a colour-coded legend distinguishing algebraic validity (green tick) from contextual validity (blue tick).	Say: 'Your Solution Pathway Model shows every step from phenomenon to answer. Today we completed the model. Add Steps 5 and 6 now — use the exact language from our three-question Validity Decision Rule.' Display the step labels on the board: 'Step 5: Verify — substitute each root; confirm both sides equal zero. Step 6: Decide — apply the three context questions; accept or reject each root with a written reason.' Allow 4 minutes for learners to update their models. Circulate and check that Step 5 includes actual substitution arithmetic (not just a label) and that Step 6 includes a written sentence, not just a tick or cross. Say to the class: 'When you show your model to someone who was not in this classroom, they should be able to follow every step and understand why Achieng's answer is $r = 8$ rows, giving $8 \times 15 = 120$ seats. Make your model that clear.' Close the lesson: 'Next lesson, Lesson 7, we will explore what happens when factorisation is difficult or impossible — but every step in today's model will still apply.'	Cumulative Model Revision (Progressive Diagram Building): the model functions as a visible, growing artefact of the entire learning sequence. Adding Steps 5 and 6 today completes the logical chain from phenomenon to solution, giving learners a portable reference tool and reinforcing that solving a quadratic equation is not finished at the factorisation step — verification and contextual interpretation are mathematically obligatory. The annotated mini-example embeds procedural fluency inside the model.	Review 4–5 models during circulation. Check: (1) Is Step 5 illustrated with actual substitution arithmetic, or is it just a label? (2) Does Step 6 include a written contextual reason (e.g., 'width cannot be negative because it is a physical dimension'), or only a tick/cross? (3) Is the overall model coherent — can a reader follow the chain from Lesson 1's phenomenon all the way to today's verified answer? Flag models that are incomplete or logically disconnected for individual follow-up at the start of Lesson 7.
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D. TEACHER REFLECTION

1. Did all learners demonstrate the distinction between algebraic validity (the substitution balances) and contextual validity (the value makes physical sense)? Which specific problems exposed gaps in this distinction, and how will I address these in Lesson 7?
2. During the peer-audit swap, what was the most common error learners circled in their partner's work — was it an arithmetic sign error, a failure to set the substitution equal to zero, or a missing contextual verdict? How does this inform the warm-up activity for Lesson 7?
3. Did learners find Problem 4 (Iten athletics, both roots valid) genuinely surprising? If most defaulted to rejecting one root without reasoning, does the class need another 'both roots valid' example before progressing to the quadratic formula in later lessons?
4. How many sticky notes were moved to the ANSWERED section of the DQB today, and how many new questions were posted? Do the new questions indicate conceptual curiosity (e.g., 'Can a quadratic have no solutions?') or persistent confusion (e.g., 'I still don't understand why we set it to zero?')?
5. Were learners' cumulative Solution Pathway Models complete enough to serve as a genuine study reference — including both substitution arithmetic and written contextual verdicts — or are they still predominantly symbolic without prose explanation? What scaffolding will help the latter group before the end-of-sub-strand

assessment?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed that substituting $r = 8$ into $r^2 + 7r - 120 = 0$ gives $64 + 56 - 120 = 0$ ✓, and substituting $r = -15$ gives $225 - 105 - 120 = 0$ ✓ — both roots satisfy the equation algebraically. They also observed that in the Kisumu market-trader problem, one root was negative (contextually impossible as a physical width), while in the Iten athletics problem both roots were non-negative and both could be physically meaningful.
What did I learn?	Verifying a solution means substituting it back into the original equation and confirming the result equals zero — this is a separate, obligatory step after factorisation. A quadratic equation always yields two algebraic roots, but the real-life context determines how many are valid: sometimes only one, sometimes both, and (in cases not yet encountered) possibly neither. The three-question Validity Decision Rule — checking sign constraints, whole-number constraints, and physical possibility — is the tool for making this judgement systematically.
How does this explain the phenomenon?	Achieng's original puzzle is now fully resolved: both $r = 8$ and $r = -15$ are genuine solutions to $r^2 + 7r - 120 = 0$, which is why the equation is correct and why 'guessing and checking' failed her. However, only $r = 8$ is valid for the chair arrangement at the Kisumu prize-giving ceremony, because the number of rows cannot be a negative number. The 8 rows and 15 columns ($8 + 7 = 15$) give exactly $8 \times 15 = 120$ seats, confirming the mathematical model matches the real-world situation perfectly.

LESSON 7 (80 minutes): Applications and Digital Research: Quadratic Equations in the Real World

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners use digital devices or print resources to research real-world applications of quadratic equations — in architecture, agriculture, sports, and business — observing online safety measures. Small groups prepare short presentations linking their findings to the driving question. The teacher facilitates a gallery walk and class discussion. Learners add new connections to the DQB and revise their models of 'what a quadratic equation can tell us.'
Knowledge	Learners will know that quadratic equations model real-life situations in architecture (arch design, roof spans), agriculture (rectangular farm plots, irrigation channels), sports (trajectory of a ball), and business (profit-loss optimisation); and that the solutions of a quadratic equation correspond to meaningful (or non-meaningful) answers depending on context.
Skills	Learners will be able to: (i) use digital devices or print resources to research applications of quadratic equations while observing online safety measures; (ii) identify the quadratic relationship in a real-world situation and write the corresponding equation; (iii) prepare and deliver a short group presentation linking research findings to the driving question; (iv) critically evaluate which solution(s) of a quadratic equation are valid in a given real-life context.
Attitudes	Learners will appreciate that quadratic equations are powerful tools for solving practical problems across many fields in Kenya and beyond, and will demonstrate respect, humility, and digital responsibility while collaborating and sharing findings with peers.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	Achieng and her classmates have already formed and solved quadratic equations (Lessons 1–6). Now they ask: beyond the prize-giving chair arrangement, where else do people in Kenya use this mathematics? By researching architecture, agriculture, sports, and business, learners discover that the squared-unknown structure they puzzled over in Lesson 1 appears everywhere — from the curved roof of a Kisumu community hall to the optimal maize-planting layout on a Rift Valley farm. This lesson expands the driving question from 'how do we solve it?' to 'why does this equation keep appearing, and what does each solution tell us about the real world?'
Safety Notes	Learners must observe online safety measures at all times: use only school-approved websites or the Kenya Education Cloud; do not share personal information online; report any inappropriate content to the teacher immediately. If physical devices are unavailable, pre-printed research cards are provided as an alternative.

B. LESSON OVERVIEW

In this 80-minute lesson, learners work in small groups (4–5) to research real-world applications of quadratic equations across four domains — architecture, agriculture, sports, and business — using digital devices or print resource cards. Each group prepares a short poster-style presentation (or digital slide) linking their findings to the driving question. A structured gallery walk allows all groups to share evidence, after which the teacher facilitates a whole-class sensemaking discussion. Learners update the Driving Question Board (DQB) with new application connections and revise their cumulative models to reflect what quadratic equations reveal in each context. The lesson is anchored throughout to the Kisumu prize-giving phenomenon: just as Achieng's chair-arrangement problem produced $x(x+7)=120$, learners discover that the same structural pattern — a product of two related linear expressions equalling a known quantity — recurs across Kenyan real-life situations.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Creativity break: How do you apply creativity in algebra Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners individually write answers to two prompt questions in their exercise books (3 minutes silent writing): (1) 'Besides arranging chairs for Achieng's prize-giving ceremony, name ONE situation in Kenya where you think a quadratic equation might be useful — and explain why you think a squared term would appear.' (2) 'If solving a quadratic equation gives you two answers, say $x = 8$ and $x = -15$, what would you do with those two answers in a real-life problem?' Learners then share predictions with a partner (Think-Pair) for 2 minutes before selected pairs share with the class.	Post the two prompt questions on the board before learners enter. Say: 'Before we research anything today, I want to know what YOU already think. Write your two answers silently — do not discuss yet. You have 3 minutes.' [Start timer. Maintain silence. After 3 minutes:] 'Now turn to your partner and share. You each have 1 minute.' [After pair share:] Cold-call FOUR pairs (aim for gender and ability mix) using the prompt: 'Wanjiku, what situation did you predict, and why does a squared term appear there?' Apply 10-second WAIT TIME before accepting any response. Record key predictions on the board under the heading 'Our Predictions — Before Research.' Do NOT confirm or correct yet — simply say: 'Interesting. Let's see what the evidence tells us.'	Think-Pair-Share with Prediction Recording — activates prior knowledge from Lessons 1–6, surfaces existing mental models, and creates a public 'prediction wall' against which post-research findings will be compared. This makes conceptual change visible and motivates the research phase.	Observable indicator: Each learner has written at least one plausible real-life context (architecture, farming, sports, business, etc.) and has articulated — however informally — why a product of two related quantities produces a squared term. Listen for learners who say only 'I don't know' — pair these with stronger predictors and note for targeted support during group research.
Observe Phase  VIDEO: Creativity break: How do you apply creativity in algebra Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners work in four home groups (assigned before the lesson), each researching one domain. Group A: Architecture (e.g., arch spans in Nairobi's KICC building, rectangular floor plans). Group B: Agriculture (e.g., optimal rectangular plot dimensions for a maize shamba in the Rift Valley, irrigation channel cross-sections). Group C: Sports (e.g., trajectory of a football kicked by Harambee Stars players at Kasarani Stadium — projectile motion producing $h = -5t^2 + 20t + 1$). Group D: Business (e.g., a Mombasa market trader's profit function $P = -2x^2 + 80x - 300$ where x is the number of units sold). Each group uses digital devices (Kenya Education Cloud / approved sites) OR a pre-printed	Distribute research cards and confirm device access (2 min). Say: 'Each group has a different real-world domain. Your job is to find a genuine example, write the quadratic equation it produces, solve it, and decide which answer makes sense in real life — just like Achieng had to choose between her two answers for the chairs. You have 20 minutes to research, then 8 minutes to prepare your poster. Online safety rule: only approved sites or the Kenya Education Cloud. If you see anything inappropriate, tell me immediately.' [Circulate every 4–5 minutes. At each group, ask ONE probing question — see below — then move on.] Probing questions by group: Group A (Architecture):	Jigsaw Research with Domain Specialisation — each group becomes the class 'expert' on one application domain. This distributes the cognitive load of exploring multiple contexts while ensuring all learners receive full breadth of applications through the gallery walk that follows. Anchoring each domain to a Kenyan context (Kasarani, Rift Valley shamba, Mombasa market) keeps the research relevant and the phenomenon central.	Observable indicators during circulation: (1) The group's written equation is correctly formed (quadratic, not linear). (2) At least one group member can explain verbally why the squared term arises (product of two related unknowns). (3) The group has identified which solution is not valid in context (e.g., negative length, negative time) — this is the key conceptual target for this lesson. Flag any group whose equation is linear or whose poster does not address solution validity for whole-class discussion in Phase 3.

	research card (provided by teacher) to: (i) find or read one concrete example of their domain; (ii) identify the quadratic equation in that example; (iii) write down what the two solutions mean in context; (iv) create a poster or digital slide with: the real-life situation described in one sentence, the quadratic equation written clearly, the solutions found (by factorisation or other method from prior lessons), and a statement about which solution is valid and why. Groups have 20 minutes for research and 8 minutes for poster/slide preparation.	'If the width of the arch is w metres and the length exceeds the width by 5 m, and the area is 84 m^2, what equation does that give you?' Group B (Agriculture): 'Farmer Wambua has a rectangular shamba. The length is 3 m more than the width and the area is 180 m^2. Write the equation for the width.' Group C (Sports): 'At what time does the ball hit the ground? What do you do with the negative time answer?' Group D (Business): 'At what number of units sold is the profit zero? Which of the two values makes business sense?' At 25 minutes: give a 3-minute warning: 'Three minutes left for research — move to poster preparation.' Remind groups: 'Your poster must show the equation clearly — a person walking past must be able to read and understand it without you explaining.' At 5 minutes remaining in poster prep: 'One minute — finalise your title and the sentence about which solution is valid.'		
Explain Phase  VIDEO: Creativity break: How do you apply creativity in algebra Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES	Posters are displayed around the room (or slides projected in rotation). Learners conduct a structured gallery walk (10 minutes): each group visits the other three posters, leaving one 'star' sticky note (something correct and clear) and one 'arrow' sticky note (a question or suggestion). After the gallery walk, learners return to their seats. The teacher leads a 15-minute whole-class discussion synthesising findings across all four domains. Learners record in their exercise books: 'The quadratic structure	Gallery Walk Setup (1 min): 'Each group will visit the other three posters. Leave a STAR sticky note — one thing that is mathematically correct and clearly explained. Leave an ARROW sticky note — one question you have OR one thing that could be improved. You have 10 minutes. Move quietly and read carefully.' [Time the gallery walk strictly. At 9 minutes: '1 minute remaining — return to your posters and read the notes left for you.'] Whole-class discussion (15 min): Stand at the board with the prediction list from Phase 1. Say:	Gallery Walk with Star-Arrow Protocol followed by Socratic Whole-Class Discussion — the gallery walk externalises peer assessment and surfaces misconceptions before whole-class discussion. The Socratic sequence ($Q1 \rightarrow Q2 \rightarrow Q3$) moves learners from noticing surface similarities across examples, to understanding why extraneous solutions arise, to constructing a general rule. This mirrors the NGSS practice of constructing explanations from evidence.	Observable indicators: (1) During gallery walk, arrow sticky notes reveal genuine mathematical questions (not just spelling corrections) — collect these for Q&A. (2) During Q1 discussion, learners use the phrase 'product of two related expressions' or equivalent — if absent, re-prompt. (3) For Q3, learners can state: 'We reject a solution if it gives a negative length, area, time, or quantity, because these have no physical meaning in the problem.' Learners who cannot connect to the phenomenon (Achieng's $x =$

for similar readings	that appeared in Achieng's chair problem (rows $\times$ columns = 120, where columns = rows + 7) also appears in [architecture / agriculture / sports / business] because ____.' Learners then answer the critical question: 'In Achieng's problem, one solution was $x = 8$ (rows) and the other was $x = -15$. Why did we reject $x = -15$? Give a parallel example from today's research where one solution is rejected — and state why.'	'Let us now compare what we predicted with what the research showed.' Point to each prediction and ask: 'Did the evidence support this, extend it, or challenge it?' Cold-call SIX learners across the four groups. After each response, apply 12-second WAIT TIME before commenting. Key questions to pose in sequence: (Q1) 'What is the ONE thing all four examples have in common mathematically?' [Target: a product of two related linear expressions equals a known constant, producing $x^2 + bx + c = 0$.] (Q2) 'Group C, your sports example gave a negative time as one solution. Group A, your architecture example gave a negative length. Why does mathematics produce these answers if they are not physically possible?' [Target: the equation describes a broader mathematical relationship; context filters valid solutions.] (Q3) 'Connect back to Kisumu: Achieng rejected $x = -15$ rows. Based on today's research, write a general rule for when we reject a quadratic solution in a real-life problem.' Write learner-generated rule on the board verbatim, then refine collaboratively. Say: 'This rule — rejecting solutions that contradict physical or contextual constraints — is one of the most important mathematical habits of mind you will use throughout your lives.'		-15) need targeted support — note their names for follow-up in Phase 5.
Driving Question Board (DQB) Creation  VIDEO: Creativity break: How do you apply creativity in	Learners spend 8 minutes updating the class DQB. Each group adds: (i) one new APPLICATION card (domain + Kenyan context + equation structure, e.g., 'Agriculture — Rift Valley maize shamba: $w(w+3) = 180$'); (ii) one ANSWER card that	Direct learners to the DQB: 'You have 8 minutes. Each group adds three cards — an APPLICATION card, an ANSWER card for something we wondered before, and a NEW QUESTION card your research raised. Write clearly so the whole class can read from a	Public Knowledge-Building on the DQB — making individual and group thinking visible on the DQB creates a shared class knowledge artefact that tracks conceptual growth across the entire 8-lesson sequence. The Before/After reflection prompts metacognition,	Observable indicators: (1) APPLICATION cards contain a correctly structured equation (not just a description). (2) ANSWER cards cite specific evidence from today's research ('Group B found that...'). (3) NEW QUESTION cards are mathematically

algebra Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	responds to a previously posted question on the DQB (from Lessons 1–6) if their research provides evidence for it; and (iii) one NEW QUESTION card that their research raised (e.g., 'Can quadratic equations model situations where the answer is never a whole number?'). Individually, each learner also writes in their exercise book: 'Before Lesson 7, I thought quadratic equations were only useful for _____. Now I know they also apply to _____ because _____.'	distance.' [Circulate and read cards as they go up. If an application card lacks an equation, prompt: 'Can you add the equation structure next to your context?'] After 8 minutes, read aloud THREE new application cards and ONE new question card, chosen to represent different domains. Say: 'Notice how the DQB is growing — when Achieng first drew her chair layout in Lesson 1, we had one question. Now we have evidence from architecture, agriculture, sports, and business. Our model of what a quadratic equation can do is getting much bigger.' Point to the Before/After reflection in learners' books: 'This reflection is evidence of YOUR thinking changing. That is what learning looks like.'	helping learners recognise their own conceptual change, a key CBE competency (learning to learn).	grounded, not vague. (4) Before/After reflections show a genuine expansion of scope (from one context to multiple contexts). Collect exercise books of three learners (one from each performance band) to read Before/After reflections as informal diagnostic data before Lesson 8.
Model Building Phase  VIDEO: Creativity break: How do you apply creativity in algebra Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	Learners individually revise their cumulative model (started in Lesson 1 and updated in each subsequent lesson). Today's revision must add: (a) at least TWO new real-life domains where quadratic equations appear, with a labelled arrow showing how the domain connects to the general form $ax^2 + bx + c = 0$; (b) a 'solution validity filter' box — a visual reminder that real-life constraints (no negative lengths, areas, times, or quantities) determine which mathematical solution is physically meaningful; and (c) a revised answer (or a more nuanced answer) to the driving question written at the top of the model page: 'How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?' Learners have 10	Say: 'Open your cumulative models. In Lessons 1 to 6 you built a model of what a quadratic equation is and how to solve it. Today you discovered WHERE it lives in the real world. Add at least two domains with arrows to the general form, add a solution-validity filter box, and write a better answer to our driving question at the top. You have 10 minutes — work in silence.' [Circulate for first 5 minutes, noting learners who are not adding new domains — prompt: 'Which of the four groups' examples could YOU add to your model? Choose one and draw the arrow.'] At 10 minutes: 'Two volunteers — who will share?' Cold-call if no volunteers. For each model shared, facilitate class feedback: 'One warm — something strong about this model. One cool — one suggestion.' Record the driving-	Cumulative Model Revision with Peer Feedback — iterative model revision is an NGSS Science and Engineering Practice (Developing and Using Models) adapted for mathematics. Each revision cycle makes the model more complete and more connected. The solution-validity filter box targets the key conceptual hurdle of this lesson: that mathematics produces solutions the real world must judge. Peer warm/cool feedback develops communication and collaboration competencies mandated by KICD.	Observable indicators: (1) Model contains labelled arrows from at least two new real-life domains to the general quadratic form. (2) Solution-validity filter box is present and contains at least one example (e.g., 'negative length → reject'). (3) Revised driving-question answer explicitly mentions multiple domains and the idea of choosing between solutions based on context. (4) Learners who produced models with no new domains or no filter box are flagged for targeted review at the start of Lesson 8.

	minutes. At the end, two volunteers share their revised models under the visualiser or by holding them up, and the class gives one piece of 'warm' and one piece of 'cool' feedback per model.	question answers written by both volunteers on the board. Say: 'Compare these two answers to what Achieng's classmates said in Lesson 1. Has your class's understanding of the driving question grown? I think it has. In Lesson 8, we will bring everything together for our final response.' End with: 'Achieng's chair problem was one arrangement in Kisumu. Today you showed it is also a roof in Nairobi, a shamba in the Rift Valley, a goal kick at Kasarani, and a market stall in Mombasa. That is the power of algebra.'		
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D. TEACHER REFLECTION

After the lesson, reflect on the following questions: (1) Did all four groups produce a correctly formed quadratic equation from their research domain — or did any group produce a linear equation instead? If so, what misconception drove that, and how will you address it in Lesson 8's consolidation? (2) During the gallery walk, what did the arrow sticky notes reveal about gaps in learners' understanding of solution validity? Were learners able to articulate WHY a negative solution is rejected, or did they simply say 'it doesn't make sense'? (3) How effectively did learners connect their research findings back to the Kisumu chair-arrangement phenomenon? If the connection was weak, consider opening Lesson 8 with an explicit comparison table: Achieng's problem vs. each of the four research domains, column by column. (4) Review the Before/After reflections from the three exercise books collected. Do learners demonstrate genuine conceptual expansion, or are they restating the same single context? Use this to calibrate the depth of consolidation needed in Lesson 8. (5) Was the 80-minute timing sufficient? If the gallery walk ran over, consider reducing Phase 1 (Predict) to 4 minutes in future by posting prompt questions as a warm-up task before class begins.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners researched and identified real-world examples of quadratic equations in architecture (e.g., rectangular floor plans and arch spans in Nairobi), agriculture (e.g., rectangular maize shamba dimensions in the Rift Valley), sports (e.g., football trajectory at Kasarani Stadium), and business (e.g., a Mombasa market trader's profit function). They observed that in each case, a product of two related linear expressions — equal to a known total — generates a quadratic equation of the form $ax^2 + bx + c = 0$.
What did I learn?	Learners learned that: (i) the quadratic structure first seen in Achieng's chair problem (rows $\times$ columns = 120, where columns exceed rows by 7) recurs across many real-life domains in Kenya; (ii) solving a quadratic equation produces two mathematical solutions, but real-life context (no negative lengths, areas, times, or quantities) determines which solution is physically valid; (iii) observing online safety measures is a responsibility when using digital devices for mathematical research; and (iv) quadratic equations are a tool for decision-making — in design, farming, sport, and commerce.
How does this explain the phenomenon?	Learners explained why a squared unknown appears in real-life problems: whenever two related unknown quantities are multiplied together to give a known area, total, or quantity, the relationship produces a quadratic equation. They also explained the solution-validity rule: after solving by factorisation, substitute both solutions back into the real-life context and reject any solution that contradicts physical constraints (e.g., you cannot have a negative number of rows of chairs, a negative field width, or a negative

	time). This rule was connected explicitly to Achieng's rejection of $x = -15$ in Lesson 1's phenomenon.
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LESSON 8 (80 minutes): Model Building, Consolidation & Final Explanation: The Full Journey of a Quadratic Equation

A. SPECIFIC LEARNING OUTCOMES

Category	Specific Learning Outcomes
Purpose	Learners consolidate the entire sub-strand by constructing a cumulative visual model (poster or digital slide) that maps the full journey from real-life context → algebraic expression → quadratic expression → quadratic equation → factorisation → solution → interpretation in context. They present models, close the Driving Question Board, complete a written assessment, and articulate how quadratic equations are used across careers in Kenya.
Knowledge	Learners demonstrate understanding that: (a) real-life situations involving two related unknown dimensions whose product is known can be modelled by quadratic equations; (b) a quadratic equation of the form $x^2 + bx + c = 0$ is solved by factorisation; (c) both solutions must be tested against the real-life context, and solutions that are negative or otherwise impossible must be rejected; (d) quadratic equations appear across Kenyan careers including architecture, agriculture, event management, and engineering.
Skills	Learners are able to: (a) form algebraic expressions and quadratic equations from real-life scenarios; (b) factorise quadratic expressions and solve quadratic equations by the factor method; (c) interpret both solutions of a quadratic equation in context; (d) construct and present a cumulative visual model showing every stage of the problem-solving journey; (e) evaluate peers' models using a structured rubric.
Attitudes	Learners appreciate that mathematics is not an isolated school subject but a tool embedded in everyday Kenyan life — from arranging chairs at Kisumu prize-giving ceremonies to designing rectangular shamba plots in Kericho and planning market stalls in Gikomba. Learners display respect by listening to peers' presentations without interruption, and demonstrate responsibility by submitting an honest individual written assessment.
Key Inquiry Question	How are quadratic equations applied in real life?
Purpose in Storyline	This is the final lesson of the sub-strand. Achieng's seating-arrangement problem — which launched the unit — is fully resolved. Learners have now travelled the complete journey: they noticed a pattern that could not be solved by simple division, named it as a quadratic relationship, formed and factorised the equation, found two solutions, and interpreted which one makes sense for 120 chairs. In this lesson they make that entire journey visible in a single model, present it to the class, close every question on the Driving Question Board, and demonstrate mastery through a written class activity / assessment task.
Safety Notes	If learners use scissors or cutting tools when constructing paper poster models, they must handle sharp edges carefully and work on a stable surface. Learners using shared digital devices must observe online safety guidelines (KICD PCIs) — they must not navigate to unrelated or unsafe websites while searching for career applications of quadratic equations.

B. LESSON OVERVIEW

Lesson 8 is the culminating lesson of Sub-Strand 1.3. It opens by returning directly to the phenomenon: Mr Otieno's challenge to Achieng — arrange 120 chairs in a rectangle where columns exceed rows by 7. In the Predict phase, learners revisit their Lesson 1 predictions and compare them with what they now know. In the Observe phase, small groups examine all their completed work from Lessons 1–7 and identify the complete solution pathway displayed on a class Evidence Wall. In the Explain phase, learners construct a cumulative journey poster/digital slide in groups, then present their Final Explanation of the driving question. The DQB review follows, where every sticky note is addressed. The lesson closes with an individual written class activity / assessment task drawn from the KICD-suggested assessment rubric, covering all four ability indicators for Sub-Strand 1.3. The teacher facilitates a closing career-connection discussion linking quadratic equations to real Kenyan professions.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12  Search ARES for similar readings	Learners receive back their original Lesson 1 prediction cards (or notebooks) showing their very first attempt to solve Achieng's problem — arranging 120 chairs in a rectangle where columns exceed rows by 7. Individually (3 minutes), they read their original prediction silently. They then write two sentences on a sticky note: (1) what they predicted then, and (2) what they now know. Pairs share their sticky notes for 2 minutes before whole-class sharing.	BEFORE CLASS: Retrieve all Lesson 1 prediction cards / notebook entries and place one on each desk face-down. Display the phenomenon slide: 'Achieng at Kisumu school needs to arrange 120 chairs — columns exceed rows by 7. What are the dimensions?' Say verbatim: 'Turn over your card. Read what you wrote in Lesson 1. Do NOT change it — that is your evidence of how your thinking has grown.' WAIT TIME: Allow 3 full minutes of silent individual reading and writing. After pairs share, cold-call 3 learners (choose one who originally guessed and checked, one who left it blank, one who attempted algebra). Ask each: 'What do you know NOW that you did not know THEN?' Record three key before/after contrasts on the board in a T-chart labelled 'Lesson 1 thinking Lesson 8 thinking'. Leave this visible for the full lesson.	Before/After Prediction Comparison — Learners make their own cognitive growth visible by holding their original prediction alongside their current understanding. This activates metacognition and motivates the upcoming consolidation: they can SEE what the unit taught them.	Observable indicator: Can each learner articulate at least one specific thing they now understand that they did not in Lesson 1? Listen for phrases such as 'I now know why there is an x-squared term' or 'I know to reject the negative solution because you cannot have negative rows.' If a learner cannot articulate any growth, note them for targeted support during Phase 3.
Observe Phase  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan	Around the classroom walls, the teacher has pre-posted six labelled Evidence Stations from the unit, each with key artefacts or summary cards: Station 1 — Real-life context (Achieng's chairs, shamba dimensions, market stalls); Station 2 — Forming	Before the gallery walk, say verbatim: 'You are not tourists — you are quality checkers. At each station, your group must agree on one thing you are CERTAIN about and one thing that still puzzles you. Write both on sticky notes and post them at the station.'	Evidence Wall Gallery Walk — By moving through physical stations that sequence the unit's content chronologically, learners reconstruct the logic of the full journey. The sticky-note protocol makes thinking public and identifies gaps before the model-	Observable indicators: (a) Sticky notes at Station 3 correctly name the quadratic structure (look for 'x times x gives x-squared'); (b) Sticky notes at Station 5 correctly identify $x = 8$ as the valid solution with a real-life reason; (c) Groups that still confuse 'factorising' with

Academy (English - US curriculum) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	algebraic expressions (representing rows as x, columns as $x + 7$); Station 3 — Forming quadratic equations ($x(x+7)=120 \rightarrow x^2+7x-120=0$); Station 4 — Factorisation method (finding factor pairs of -120 that sum to $+7$: $+15$ and -8); Station 5 — Solutions ($x = 8$ or $x = -15$) and interpretation (reject $x = -15$; rows = 8, columns = 15); Station 6 — Careers in Kenya that use quadratic equations. Groups of 4 (pre-assigned) rotate through all 6 stations in 12 minutes (2 minutes each), adding one sticky note per station: something they CONFIRM they understand and one question they still have.	Set a timer visible to all. Call 'Rotate!' every 2 minutes with a clap signal. Circulate continuously. At Station 3 (forming the equation), pause with each group and ask: 'Why is the equation NOT just $7x = 120$? What changes when the unknown appears in BOTH dimensions?' WAIT TIME: 12 seconds. Cold-call 2 groups to respond aloud. At Station 5 (interpretation), ask: 'Achieng found $x = 8$ and $x = -15$. She only used one. Which one and why?' WAIT TIME: 10 seconds. Cold-call 3 learners by name. After the walk, spend 3 minutes harvesting 'still puzzled' sticky notes. Read 3 aloud and tell the class: 'We will answer all of these before this lesson ends.'	building phase. This mirrors the NGSS Science and Engineering Practice of 'Obtaining, Evaluating, and Communicating Information.'	'expanding' are noted — teacher provides a targeted prompt card at their table before Phase 3 begins.
Explain Phase VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum) Search ARES for similar videos HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	Each group (same groups of 4 from Phase 2) constructs a Cumulative Journey Model — a poster (A2 paper with markers) or a 6-slide digital presentation on a shared device. The model must include ALL seven labelled stages with arrows showing the direction of reasoning: (1) Real-life Context $\rightarrow$ (2) Identifying the Unknowns $\rightarrow$ (3) Forming the Algebraic Expression $\rightarrow$ (4) Forming the Quadratic Equation $\rightarrow$ (5) Factorising the Expression $\rightarrow$ (6) Solving for x $\rightarrow$ (7) Interpreting the Solution in Context. Each stage must include: the mathematical notation, a sentence in plain English, and a Kenya-relevant example from the unit (Achieng's	Distribute large A2 paper or ensure each group's device is open to a blank slide deck. Say verbatim: 'Your model must tell the FULL story — from the moment Achieng could not solve her problem by guessing, all the way to the moment she told the head teacher: 8 rows, 15 columns. Every arrow matters. Every stage needs BOTH the algebra AND the plain-English sentence.' During construction (15 minutes), circulate with two priority questions: (a) 'Show me where in your model the squared term FIRST appears — why does it appear there?'	Cumulative Journey Modelling with Peer Presentation — This is the primary sensemaking strategy for a final lesson. By constructing an artefact that makes the full reasoning chain visible, learners synthesise seven lessons of learning into a single coherent narrative. Peer scoring activates critical evaluation (NGSS Practice: Constructing Explanations). The convergence discussion builds shared disciplinary understanding of what a 'complete solution' means in mathematics.	Observable indicators during model construction: (a) Stage 4 (forming the equation) uses correct notation $x^2+7x-120=0$ and not $x+7=120$; (b) Stage 5 (factorisation) shows the correct factor pair $(+15, -8)$ with working; (c) Stage 7 explicitly states that $x = -15$ is rejected because a negative number of rows has no real-life meaning. During presentations: listen for learners who can explain WHY, not just WHAT — 'The squared term appears because we multiply rows by columns, and the unknown appears in BOTH dimensions.'

	chairs, OR a shamba plot, OR a market-stall scenario). Groups have 15 minutes to build. Each group then presents their model to the class in 3 minutes. Peer groups use a simple 3-point rubric card (provided by the teacher) to score: correctness of mathematics, clarity of the journey, and strength of real-life interpretation.	(b) 'Where does your model show that one solution was rejected — and why?' WAIT TIME: 12 seconds after each prompt before accepting responses. During presentations, stand at the back so learners present TO the class, not to the teacher. After each presentation, ask the audience: 'What is ONE thing this group explained better than your own group?' Cold-call 2 audience members per presentation. After all presentations, facilitate a 5-minute whole-class convergence discussion. Ask: 'Looking at all models — what is the ONE stage that EVERY group agrees is the most important and WHY?' WAIT TIME: 15 seconds. Record the consensus answer on the board. (Expected answer: Stage 4 — forming the quadratic equation — because without it, the rest of the journey cannot begin.)		
Driving Question Board (DQB) Creation  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum)  Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX)	The class Driving Question Board (DQB) — which has been built up across all 8 lessons — is displayed at the front. The teacher reads each sticky-note question aloud, one by one. For each question, the class votes silently: thumbs up = 'we can fully answer this now'; thumbs sideways = 'we can partially answer this'; thumbs down = 'we still cannot answer this.' For every 'thumbs up' question, a volunteer learner provides the answer in one or two sentences, which the teacher writes on a green card and sticks over the original question. For 'thumbs sideways' and 'thumbs down'	Stand beside the DQB. Say verbatim: 'Every question on this board was asked by someone in this class. Every question was a good question. Let us honour each one with a proper answer.' Read each question slowly. After the class votes, for 'thumbs up' questions say: 'Who will answer this for us in one sentence?' WAIT TIME: 10 seconds. Cold-call the learner who originally posted that question wherever possible — give them the honour of answering their own question.	DQB Closure with Collective Verification — The DQB has tracked growing understanding across the unit. Closing it publicly creates a shared record of what the class now knows and what remains open. This mirrors scientific practice: claims are revisited, tested against evidence, and either confirmed or identified as requiring further investigation. The 'Further Inquiry' strip models the disposition of a lifelong learner.	Observable indicators: (a) At least 80% of DQB questions receive 'thumbs up' votes, suggesting whole-class understanding; (b) Learners can answer the three anchor questions using correct mathematical vocabulary (product, squared term, factorisation, rejection of negative solution); (c) Any learner still uncertain about the distinction between 'forming the expression' and 'forming the equation' is noted for individual follow-up after the assessment task.

Source: CK-12 Search ARES for similar readings	questions, the teacher acknowledges them honestly and, where appropriate, notes which future topic (e.g., quadratic formula in Grade 11) will address them. The class also revisits the three overarching questions that launched the unit: 'Why does a seating problem produce a squared term?', 'How do you solve an equation where the unknown is multiplied by itself?', and 'When you find two answers, which one works — or do both?'	For the three anchor questions of the unit, call on 3 different learners per question. Prompt: 'Use the words ROWS, COLUMNS, PRODUCT, and SQUARED TERM in your answer.' For unresolved questions, model intellectual honesty: 'This is a real question that mathematics has an answer to — we will meet it again when we study the quadratic formula in Grade 11.' Write those questions on a 'Further Inquiry' strip and post it on the wall. Close the DQB review by asking the whole class: 'How has our driving question — how can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life Kenya? — been answered today?' WAIT TIME: 15 seconds. Cold-call 2 learners.		
Model Building Phase  VIDEO: Analyzing a cumulative relative frequency graph Source: Khan Academy (English - US curriculum) Search ARES for similar videos  HTML: Finding a Linear Product Using a Quadratic (PLIX) Source: CK-12 Search ARES for similar readings	PART A — Individual Written Assessment (20 minutes): Each learner completes a structured class activity / assessment task independently and in silence. The task contains four sections aligned to the KICD assessment rubric indicators: (i) Form a quadratic expression: 'A rectangular maize shamba in Kericho has an area of 180 m². The length exceeds the width by 3 m. Write an expression for the area in terms of the width w.' (ii) Form a quadratic equation: 'Using your expression, write a quadratic equation and rearrange it into the standard form $w^2 + bw + c = 0$.' (iii) Factorise and solve: 'Factorise the quadratic expression and solve the equation.' (iv) Interpret and apply: 'State which	Distribute printed or handwritten assessment task sheets. Say verbatim: 'This is your chance to show EVERYTHING you have learned across all eight lessons. Show every step — forming the expression, writing the equation, factorising, solving, and interpreting. A correct final answer with no working earns you very few marks. The JOURNEY is what is being assessed.' During the 20-minute assessment, circulate silently. Do NOT answer mathematical questions — instead, if a learner is stuck, point silently to the relevant stage of their journey model poster (still visible on the wall).	Structured Written Assessment as Final Model Revision — The assessment task is itself the final act of model building: learners apply the complete problem-solving journey independently to a NEW but structurally identical context (Kericho shamba). This confirms whether the understanding is transferable, not just recallable. The career-connections discussion connects disciplinary knowledge to identity and aspiration — a key dimension of KICD's CBE competency of 'self-efficacy' and the National Goal of promoting individual development and self-fulfilment.	Observable indicators for summative use: (a) Section (i): correct quadratic expression $w(w+3)$ or w^2+3w — no marks if learner writes $w+3=180$; (b) Section (ii): equation correctly rearranged to $w^2+3w-180=0$; (c) Section (iii): correct factorisation $(w+15)(w-12)=0$ yielding $w=12$ or $w=-15$; (d) Section (iv): valid solution stated as $w=12$ m (width), length=15 m, with explicit rejection of $w=-15$ on grounds that width cannot be negative. KICD rubric level: Exceeds Expectations = all four sections correct with explanation; Meets Expectations = all four sections correct; Approaches Expectations = any two of four correct; Below Expectations = any one of four

	solution is valid for the shamba and give its dimensions. Explain why the other solution cannot apply.' PART B — Career Connections Discussion (10 minutes): After submitting the assessment, learners participate in a structured Think-Pair-Share. Prompt: 'Name one Kenyan career where a professional would NEED to solve a quadratic equation, and describe a specific situation in that career where the equation would arise.' Pairs share for 2 minutes, then 6 pairs report to the whole class. Teacher records career examples on the board.	After assessment papers are collected, transition to Part B. Say: 'Quadratic equations are not only in Achieng's school in Kisumu. They show up wherever two related quantities multiply to give a fixed total. Think about a farmer in Kericho, an architect designing a block of flats in Nairobi, a market organiser setting up stalls in Gikomba, or a fish farmer at Lake Victoria.' For Think-Pair-Share: WAIT TIME 2 minutes for individual thinking before pairs discuss. Cold-call 6 pairs — aim for gender balance and varied career sectors. Record each career on the board. Close the lesson with a whole-class statement. Say verbatim: 'Every career on this board is a career that is open to you. The mathematics you have learned in this sub-strand is not a school exercise — it is a professional tool. Achieng solved her seating problem. What problem will YOU solve with it?'		correct.
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D. TEACHER REFLECTION

After delivering this lesson, reflect on the following four questions and record your notes in your professional journal:

- JOURNEY VISIBILITY:** When learners presented their cumulative journey models, were all seven stages present and correctly connected with arrows? Which stage was most commonly missing or incorrect? (Common gap: Stage 7 — learners solve correctly but fail to explicitly reject the negative solution in writing.) How will you address this in Lesson 1 of the next sub-strand by building the habit of always interpreting solutions in context?
- DQB CLOSURE:** Were there questions on the DQB that the class could not answer by 'thumbs up'? List them. Do these questions suggest a gap in your teaching sequence, or do they represent genuine Grade 11 content (e.g., quadratic formula, completing the square)? If a majority of the class gave 'thumbs down' to a question that should have been answered in this unit, plan a 10-minute retrieval activity at the start of the next lesson.

3. **ASSESSMENT QUALITY:** In the individual written assessment (the Kericho shamba task), how many learners reached Section (iv) — interpretation — and rejected the negative solution with a written reason? The KICD rubric indicator for 'Exceeds Expectations' requires learners to explain each step with varied examples. If fewer than 60% of learners reached this level, consider whether your Phase 3 model-building activity provided enough practice with interpretation before the formal assessment.

4. **CAREER CONNECTIONS:** Which careers did learners name in Part B? Were the responses specific (e.g., 'A fisheries officer at Lake Victoria calculates the dimensions of a rectangular fish pond to maximise area for a given fencing perimeter') or generic (e.g., 'engineers use maths')? If responses were generic, plan a 5-minute career-context starter activity at the beginning of your next sub-strand to deepen learners' awareness of mathematics in Kenyan professional life.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Students examined an Evidence Wall of six stations tracing the complete unit journey — from Achieng's 120-chair seating problem at a Kisumu school to the factorised solution and real-life interpretation. They identified which solution ($x = 8$ rows) is valid and why $x = -15$ must be rejected.
What did I learn?	A quadratic equation of the form $x^2 + 7x - 120 = 0$ arises when two related unknown dimensions (rows and columns) have a known product. It is solved by finding factor pairs of the constant term whose sum equals the coefficient of x . Both solutions must be tested in context — negative values are rejected because physical quantities such as rows and columns cannot be negative.
How does this explain the phenomenon?	Learners constructed cumulative journey models mapping the full path from real-life context → algebraic expression → quadratic equation → factorisation → solution → contextual interpretation. They presented these models, closed the Driving Question Board, completed an individual written assessment task (Kericho shamba dimensions), and connected quadratic equations to careers across Kenya including architecture, agriculture, event management, and fisheries.

DIFFERENTIATION AND INCLUSION		
Learner Need	Support Strategies	Extension Strategies
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.